

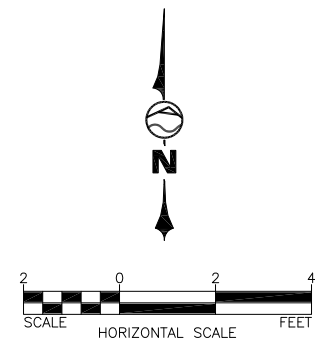
DESIGNED BY	JGC
DRAWN BY	JGS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

TRAIN 3 PLAN

SHEET	DATE
C3.3	NOV 2, 2022
	SCALE
PAGE	AS SHOWN
28 OF 96	JOB NUMBER
	2020-070



811 Call 811
two business days
before you dig

BID SET

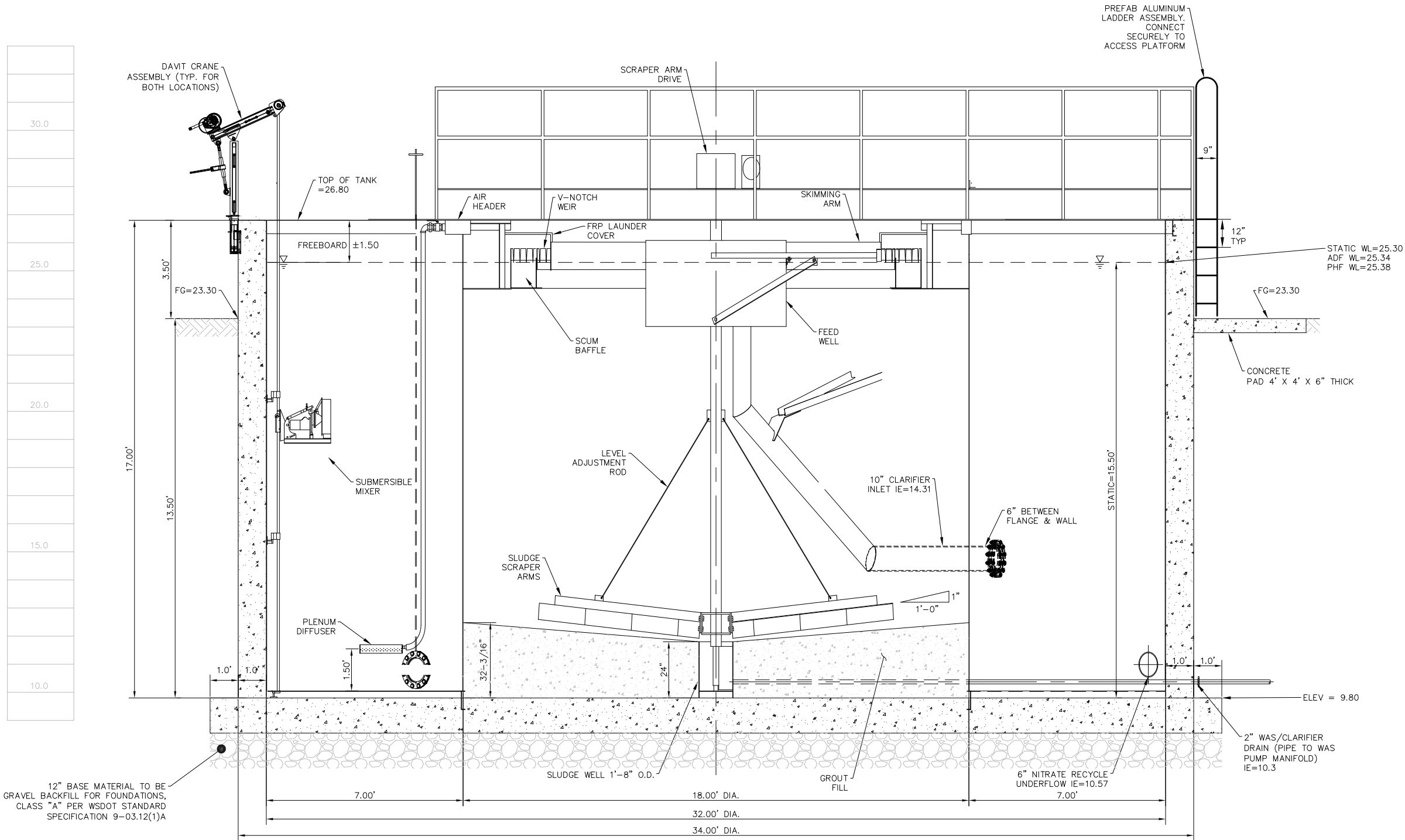
NOTES:

1. PARTITIONS TO BE REINFORCED FOR INDIVIDUAL ZONE DEWATERING. (NOT SHOWN FOR CLARITY)

PLOT SETTINGS: Adobe PDF.pc3, Ledger, Portrait, 1:2, WE APWA_UNSCREENED.ctb
W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 C3.3-C3.5 WWTP CELL 3.DWG - 11/2/2022 4:27 PM - Sam Rodriguez

NO.	REVISIONS	BY	DATE

30.0
25.0
20.0
15.0
10.0



1 SOUTH-FACING SECTIONAL ELEVATION

SCALE 0 2 4 FEET

HORIZONTAL SCALE



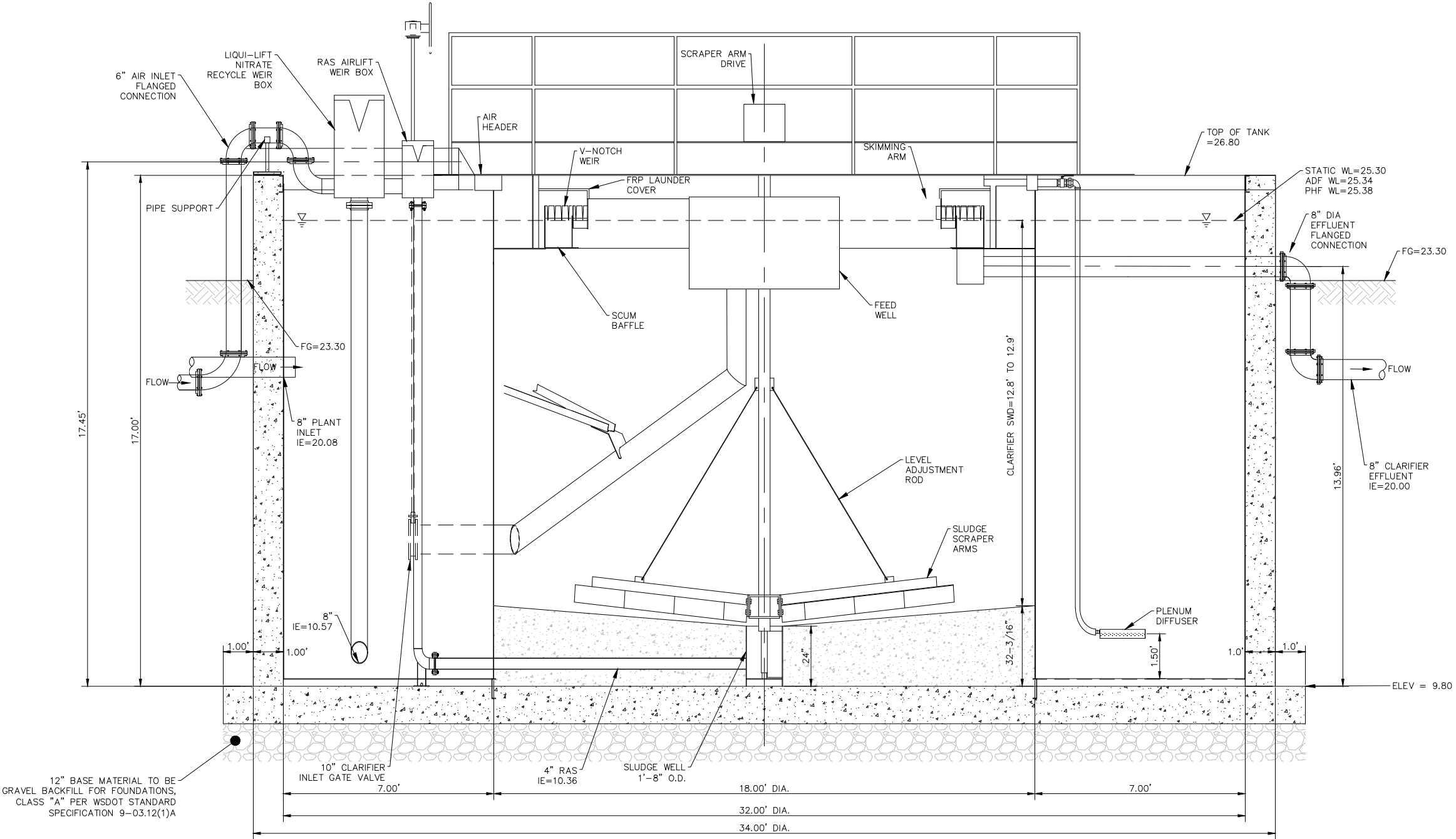
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11-02-2022

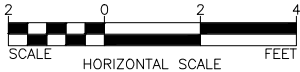
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		PAGE 29 OF 96						
EASTSOUND SEWER AND WATER DISTRICT		SAN JUAN COUNTY		WASTEWATER TREATMENT PLANT UPGRADE - PHASE I				
DESIGNED BY		JCC	DRAWN BY		JGS	CHECKED BY		AWL
WILSON ENGINEERING		G. CHRISTENSEN		WILSONENGINEERING.COM		TRAIN 3 SECTION A		

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30.0
25.0
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1 WEST-FACING SECTIONAL ELEVATION



BID SET

NO.	REVISIONS	BY	DATE

11-02-2022

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EASTSOUND SEWER AND WATER DISTRICT WASHINGTON WASTEWATER TREATMENT PLANT UPGRADE - PHASE I SAN JUAN COUNTY TRAIN 3 SECTION B	
DATE NOV 2, 2022	SCALE AS SHOWN
SHEET C3.5	JOB NUMBER 2020-070
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PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020_2020-070 C4.1-C4.3 PLANT DRAIN PUMP STATION.DWG - 11/2/2022 3:25 PM - Jeff Smith

- 1 4" D.I. SPOOL, FxMJ
2 4" D.I. SPOOL, FxF
3 4" D.I. 90° BEND, FxF
4 4" D.I. TEE, RESTRAINED
5 4" CHECK VALVE, FxF
6 4" GATE VALVE, FxF
7 4" D.I. 22.5° BEND, RESTRAINED
8 4" D.I. 45° LATERAL, FLxFL
9 4"x3" DI REDUCER (IF REQUIRED) FOR PUMP DISCHARGE, FxF
10 ROTATE TOP ELBOW 11.25° AND PROVIDE OVER-SIZED CORE HOLE, AS NEEDED, TO FIT 90° ELBOW ASSEMBLY IN WET WELL AS SHOWN
11 4" FLANGED COUPLING ADAPTOR (RESTRAINED)
12 4" D.I. PIPE, FORCE MAIN, RESTRAINED
13 PRESSURE GAGE, ISOLATOR, & SHUTOFF VALVE
14 ADJUSTABLE PIPE SUPPORTS (4 TOTAL)
15 6"x4" D.I. REDUCER (IF REQ'D.) FOR PUMP DISCHARGE, FxF
16 4" BURIED GATE VALVE, FLxMJ
17 PRECAST CONCRETE VAULT 4'-2"L x 4'-2"W x 5'-6"D (MIN. INSIDE DIMENSIONS) UTILITY VAULT CO., MODEL 506-LA, OR EQUAL
18 3" STAINLESS STEEL VENT PIPE, W/ SCREEN. IE=21.00. SCREEN TO BE 24 MESH STAINLESS STEEL.
19 TOP SLAB PER

1	2
S2.4	S2.4

20 36"x36" SINGLE LEAF HATCH, HALLIDAY PRODUCTS, H-20 LOAD RATING, MODEL H1R3636.
21 STEPS OR LADDER PER WSDOT STD. PLAN B-30.90-01. PROVIDE W/ NON-CORROSIVE MATERIALS.
22 SUSPENDED CABLE HOOKS, TYP. SEE ELECTRICAL DWGS.
23 INCOMING CONDUITS
24 GUIDE RAIL PER PUMP SUPPLIER REQUIREMENTS (TO BE STAINLESS STEEL.)
25 HAUL CHAIN BY SUPPLIER (TO BE STAINLESS STEEL)
26 PUMP POWER CORD, LENGTH TO ALLOW PUMP REMOVAL
27 VERTICAL PIPE BRACE, TYP. SEE DETAIL

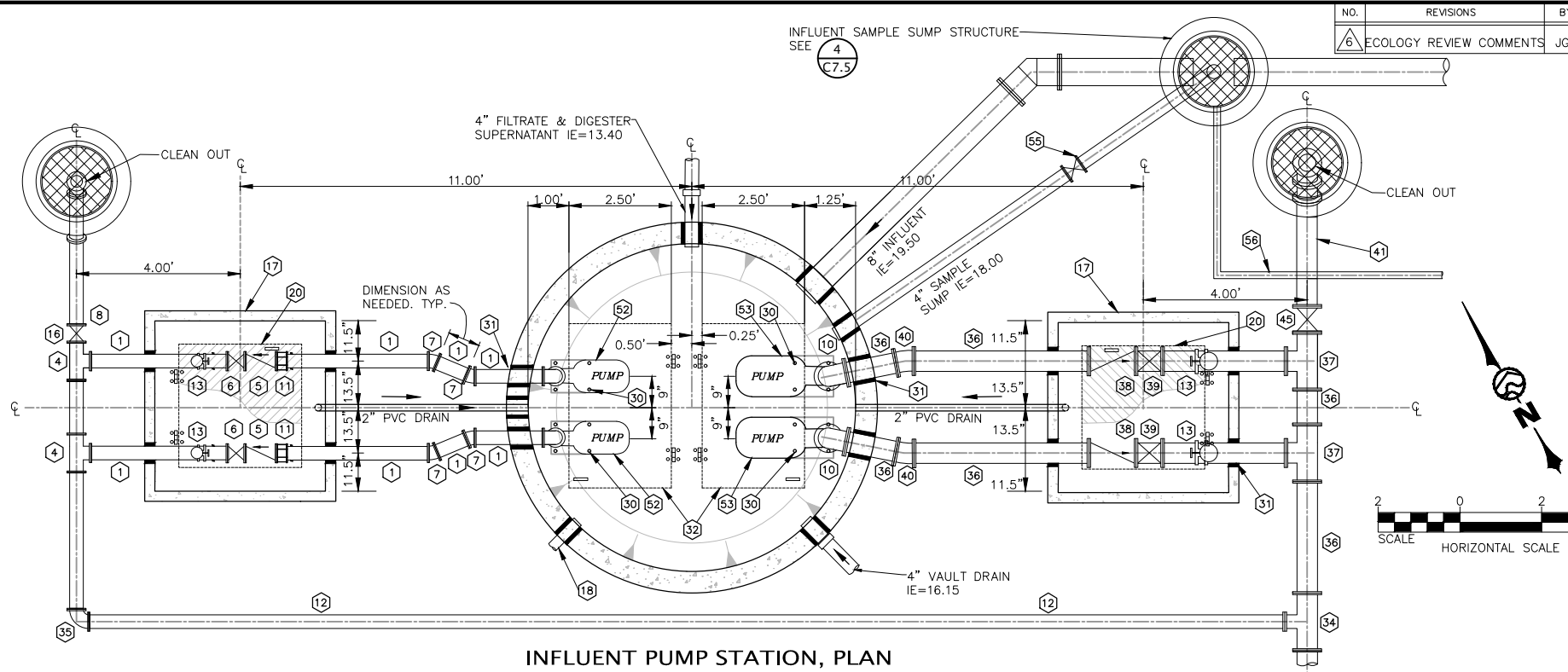
7
C7.2

28 PUMPS PER SPECS
29 MANUFACTURERS DISCONNECT ELBOW
30 3/4" S.S. ANCHOR BOLT WITH 2" PROJECTION, TYP. (16 TOTAL). LOCATE & SIZE PER MFR. REQUIREMENTS.
31 INSTALL LINK-SEAL BOOTS, OR EQUAL. TYP ON ALL PIPE-TO-CONC. CONNECTIONS. INSTALL PER MFR REQUIREMENTS. SEAL ALL VOIDS. (TYP.)
32 30"x48" SINGLE LEAF HATCH, HALLIDAY PRODUCTS, H-20 LOAD RATING, MODEL H1R3048, LOCATE HATCH OVER PUMPS COORDINATE WITH SUPPLIER
33 FLAPPER VALVE. CONFIGURE TO CLOSE & SEAL WHEN SUBMERGED
34 6"x4" D.I. TEE, RESTRAINED
35 ***VACANT***
36 6" D.I. SPOOL, RESTRAINED
37 6" D.I. TEE, FxF
38 6" CHECK VALVE, FxF
39 6" GATE VALVE, FxF
40 6" D.I. 11.25° BEND, RESTRAINED
41 6" D.I. 45° LATERAL, FLxFL
42 ***VACANT***
43 ***VACANT***
44 6" FLANGED COUPLING ADAPTOR (RESTRAINED)
45 6" BURIED GATE VALVE, FLxMJ
46 2"-4" CRUSHED ROCK
47 24"Ø X 24" LONG CONC. WELL CURB H2 PRECAST, MODEL#WC-24 W/ CONC. ADJ. RING, OR EQUAL.
48 4" x 3" PVC REDUCER & ADAPTOR AS NEEDED.
49 ANDREW'S BOSS LOCK 3" FEMALE COUPLE X MALE NPT ALUMINUM.
50 ANDREW'S BOSS LOCK TYPE DP DUST PLUG, ALUMINUM.
51 IFCO 813 RING AND COVER, NON-LOCKING
52 SMALL PUMPS DUPLEX ARRANGEMENT. SEE SPECS
53 LARGE PUMPS DUPLEX ARRANGEMENT. SEE SPECS
54 TRANSDUCER. SET 6" ABOVE FLOOR
55 4" BURIED GATE VALVE
56 2" SCH 80 CONDUIT FOR INF. SAMPLER INTAKE TUBING

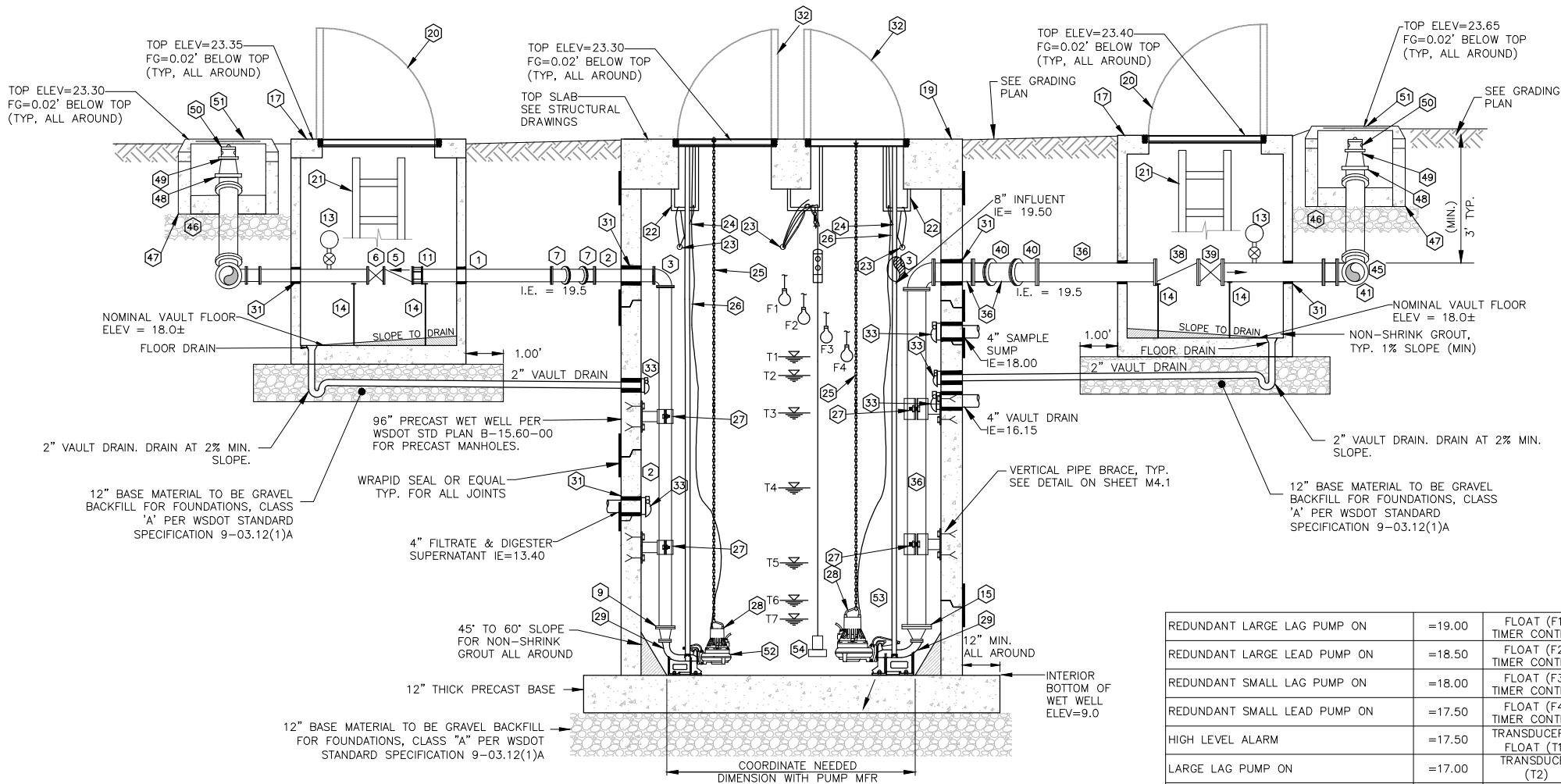


BID SET

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INFLUENT PUMP STATION, PLAN



INFLUENT PUMP STATION, ELEVATION

REDUNDANT LARGE LAG PUMP ON	=19.00	FLOAT (F1) TIMER CONTROL
REDUNDANT LARGE LEAD PUMP ON	=18.50	FLOAT (F2) TIMER CONTROL
REDUNDANT SMALL LAG PUMP ON	=18.00	FLOAT (F3) TIMER CONTROL
REDUNDANT SMALL LEAD PUMP ON	=17.50	FLOAT (F4) TIMER CONTROL
HIGH LEVEL ALARM	=17.50	TRANSDUCER & FLOAT (T1)
LARGE LAG PUMP ON	=17.00	TRANSDUCER (T2)
LARGE LEAD PUMP ON	=16.00	TRANSDUCER (T3)
SMALL LAG PUMP ON	=14.00	TRANSDUCER (T4)
SMALL LEAD PUMP ON	=12.00	TRANSDUCER (T5)
PUMPS OFF	=11.00	TRANSDUCER (T6)
LOW LEVEL ALARM & REDUNDANT OFF	=10.50	TRANSDUCER (T7)

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



11-02-2022

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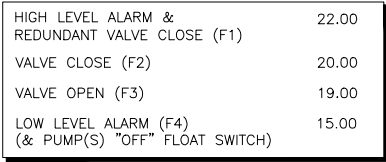
EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

SAN JUAN COUNTY

INFLUENT PUMP STATION

SHEET	DATE	SCALE	JOB NUMBER
C4.1	NOV 2, 2022	AS SHOWN	2020-070
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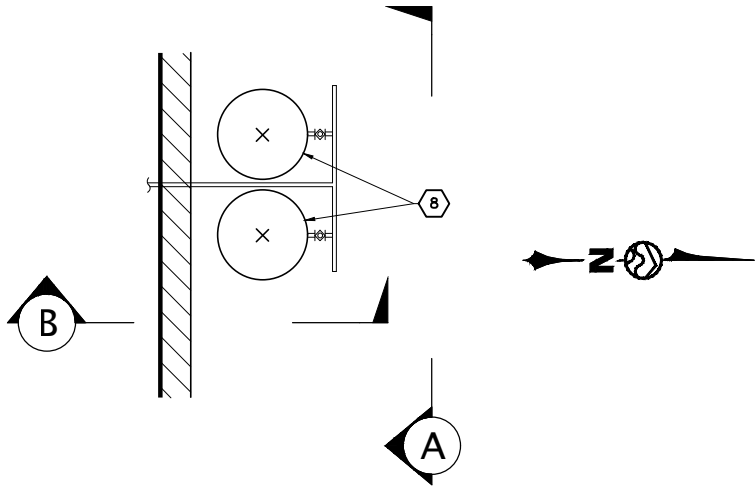
2W PLANT WATER PUMP STATION ELEVATION



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two business days
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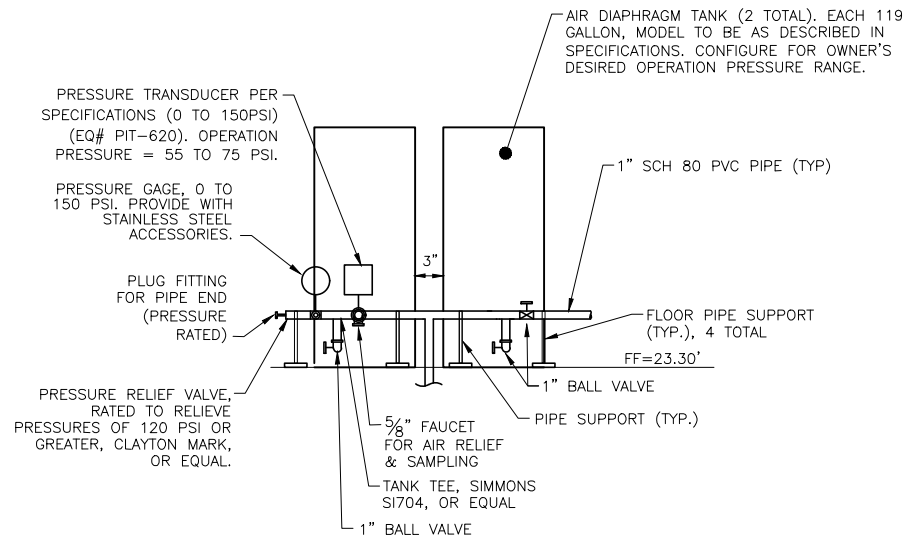
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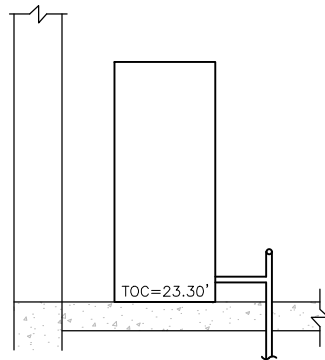
1 2W DIAPHRAGM TANKS – PLAN
PIPE SUPPORTS NOT SHOWN FOR CLARITY

2 0 2 4
SCALE HORIZONTAL SCALE FEET



A 2W DIAPHRAM TANKS – SECTION 'A'
NOT TO SCALE

- NOTES:
1. FILL ALL PIPING WITH CLEAN WATER (DOMESTIC POTABLE WATER) PRIOR TO STARTING AND TESTING 2W WATER SYSTEM EQUIPMENT.
 2. CONTRACTOR RESPONSIBLE FOR COORDINATING PIPING CONNECTION STYLE (GROOVED, FLANGED, THREADED, ETC.) AND PERFORMING PRESSURE TESTING TO DEMONSTRATE WATER TIGHT CONSTRUCTION.



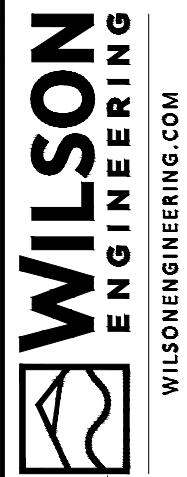
B 2W DIAPHRAGM TANK – SECTION 'B'
VALVE HANDLES AND HOIST CHAINS NOT SHOWN FOR CLARITY.



BID SET

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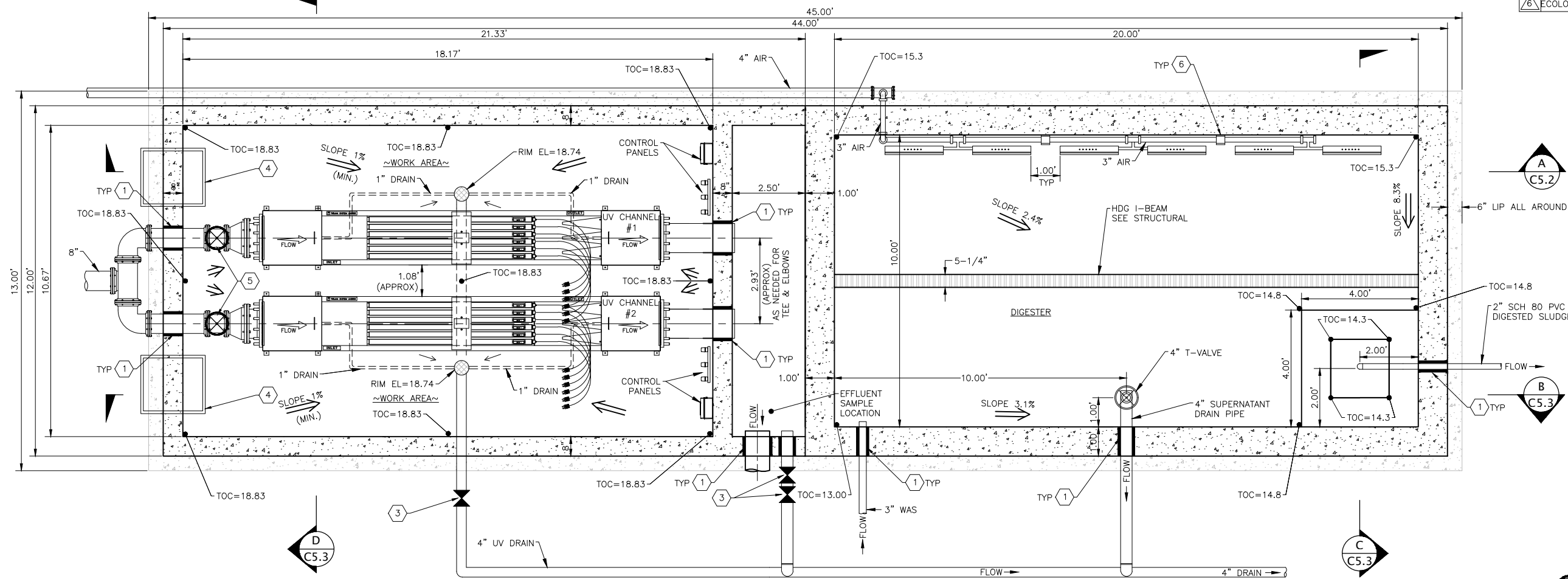
EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

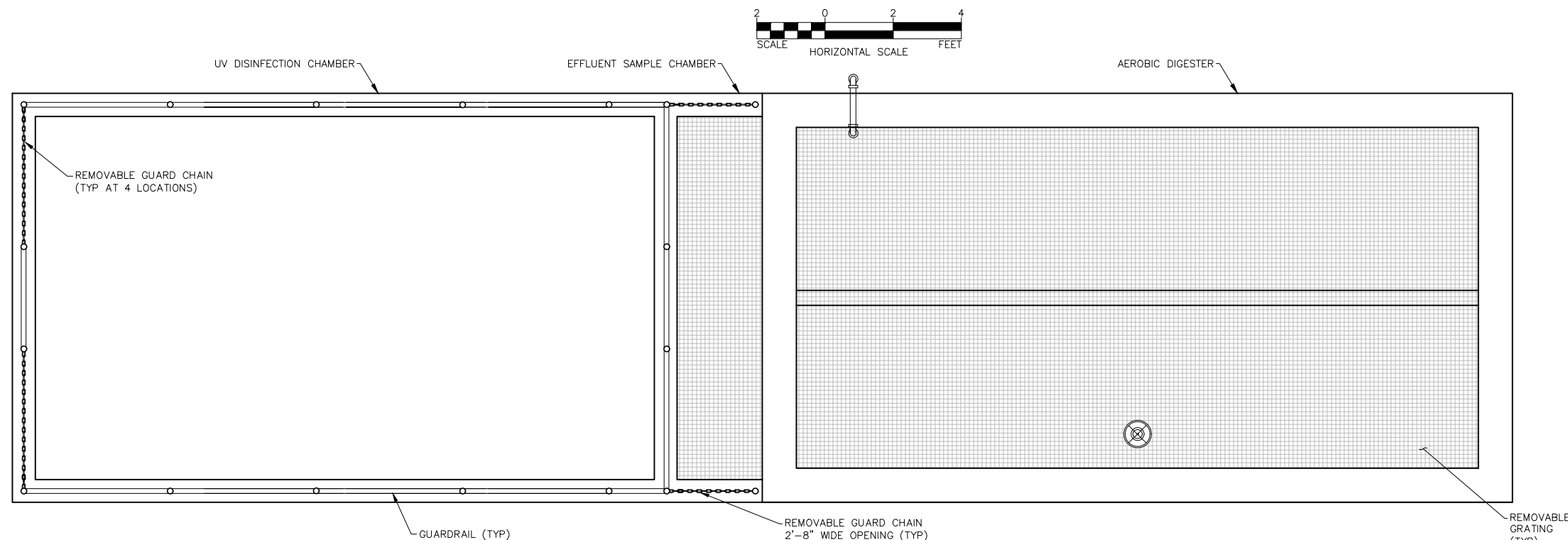
2W EQUIPMENT LAYOUT

SHEET	C4.3	DATE	NOV 2, 2022
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	OF		96

NO.	REVISIONS	BY	DATE
3	ADD UV DISINFECTION	JGC	5/10/22
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



UV DISINFECTION CHAMBER & DIGESTER – PLAN VIEW



UV DISINFECTION CHAMBER & DIGESTER – GRATING & GUARDRAIL PLAN

- KEYED NOTES
- ① = INSTALL LINK-SEAL BOOTS OR EQUAL TYP. ON ALL PIPE TO CONCRETE CONNECTIONS. SEAL ALL VOIDS.
 - ② = BRACKET, SECURELY ANCHORED TO CONC. WALL (TYP)
 - ③ = 4" GATE VALVE (TYP)
 - ④ = ALUMINUM ACCESS LADDER (NOMINAL WIDTH = 2')
 - ⑤ = 8" GATE VALVE WITH HAND WHEEL
 - ⑥ = WALL MOUNT PIPE SUPPORT PER 

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before you dig

BID SET

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3	ADD UV DISINFECTION	JGC	5/10/22
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

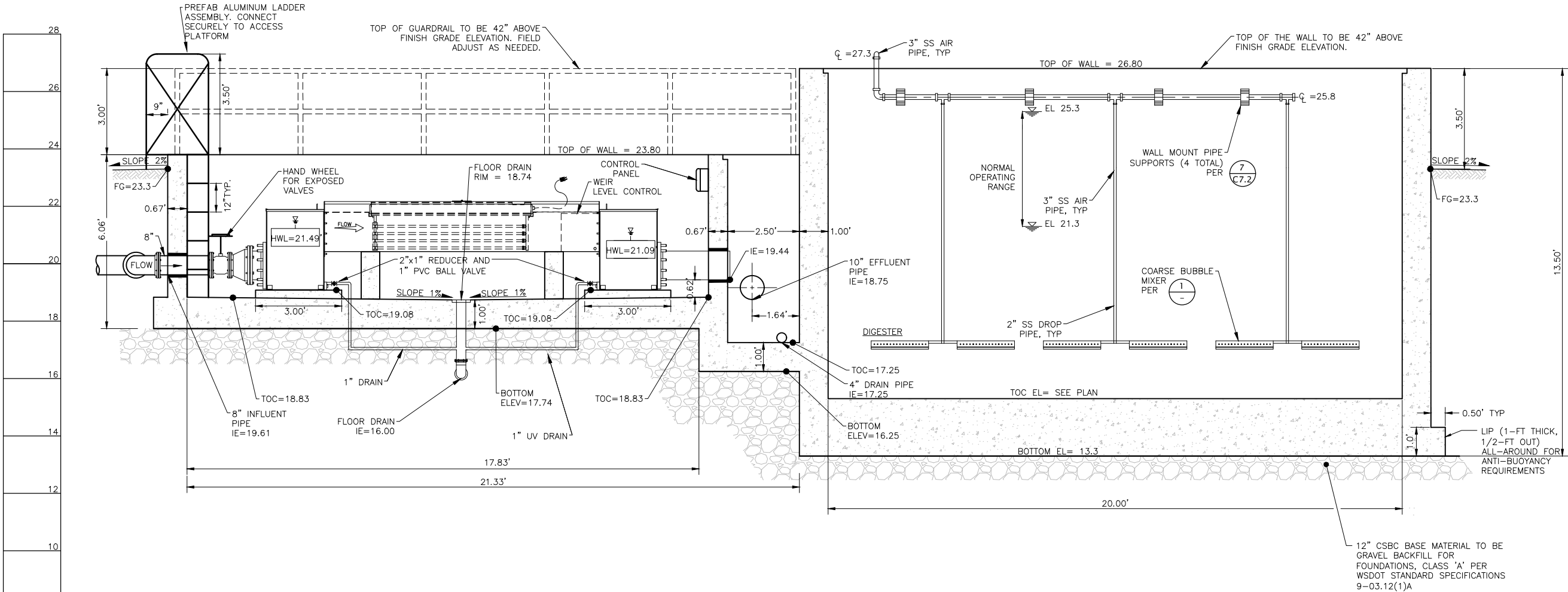
UV DISINFECTION CHAMBER & DIGESTER SECTION "A"

SHEET
C5.2

DATE
NOV 2, 2022

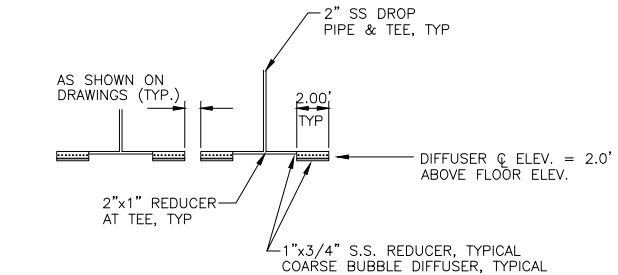
SCALE
AS SHOWN

JOB NUMBER
2020-070



A UV DISINFECTION CHAMBER & DIGESTER - SECTION VIEW 'A'

SCALE HORIZONTAL SCALE FEET



1 COARSE BUBBLE MIXER DETAIL (TYP)
NOT TO SCALE

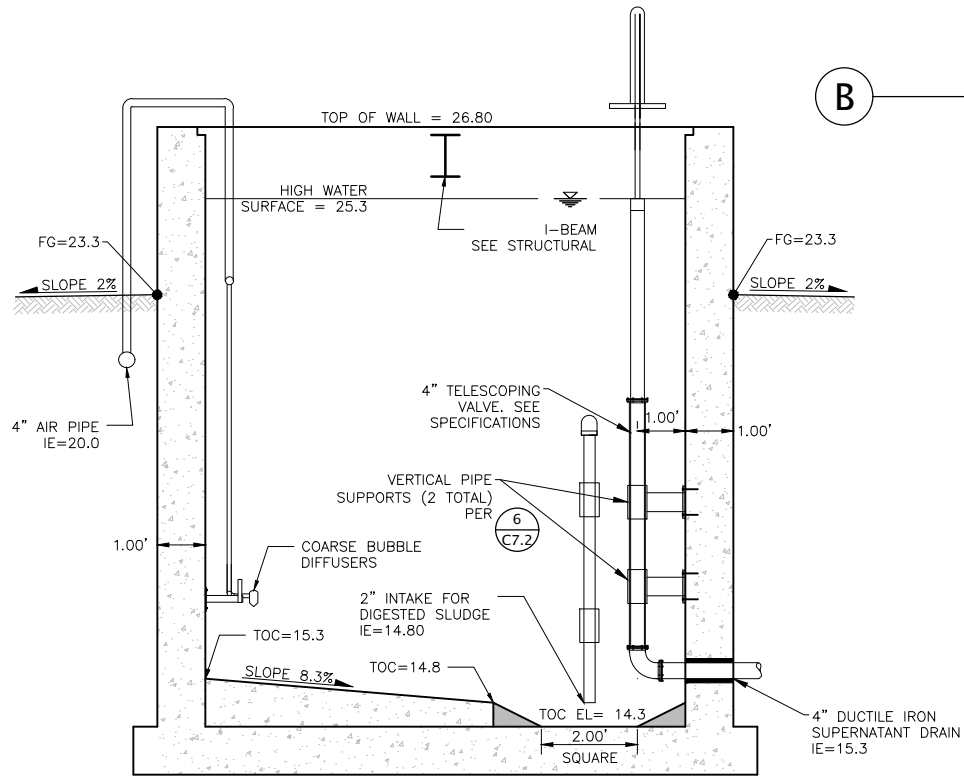
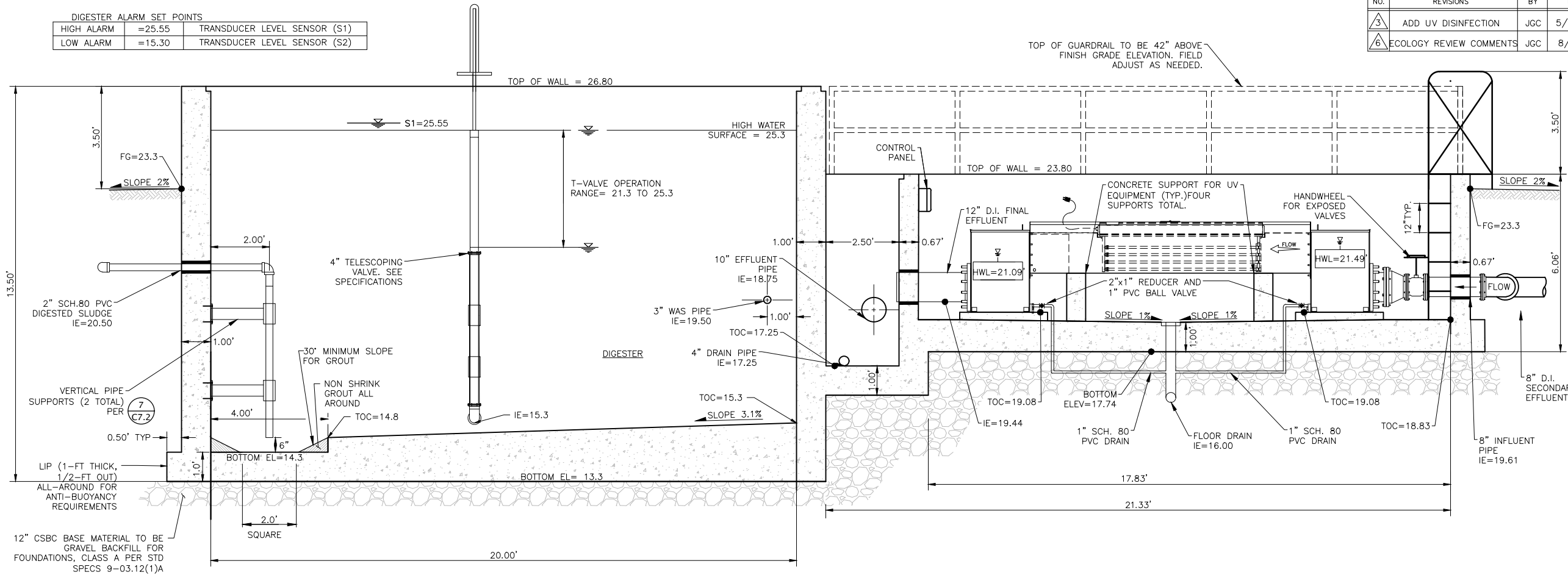


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W:\2020-2070 EASTSOUND WWTP DESIGN\DWG\2020-070 C5.1 UV DISINFECTION & DIGESTER.DWG - 11/2/2022 3:26 PM - Jeff Smith

DIGESTER ALARM SET POINTS			
HIGH ALARM	=25.55	TRANSDUCER LEVEL SENSOR (S1)	
LOW ALARM	=15.30	TRANSDUCER LEVEL SENSOR (S2)	

NO.	REVISIONS	BY	DATE
3	ADD UV DISINFECTION	JGC	5/10/22
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



DIGESTER - SECTION VIEW 'C'

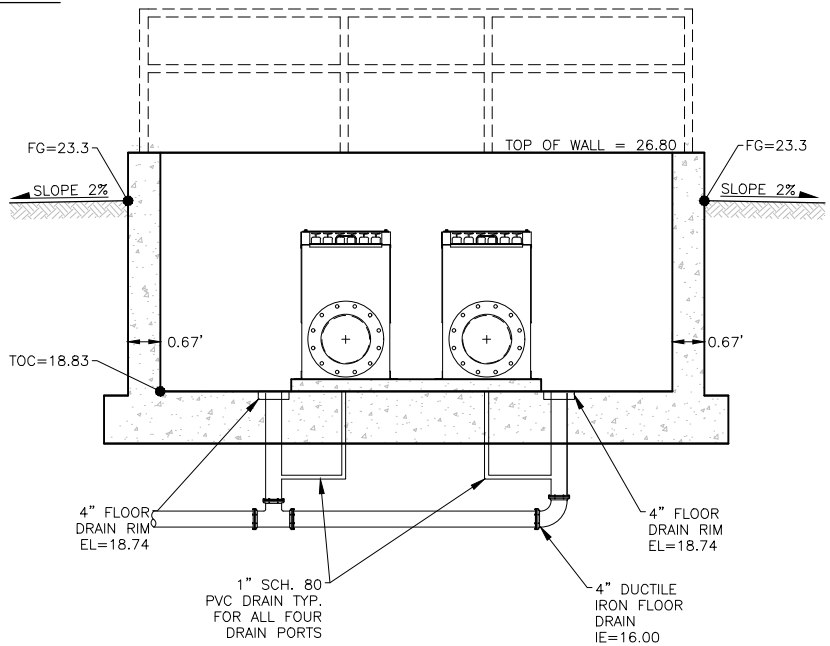


UV DISINFECTION & DIGESTER - SECTION VIEW 'B'



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two business days
before you dig

BID SET



UV DISINFECTION CHAMBER - SECTION VIEW 'D'



11-02-2022



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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

UV DISINFECTION CHAMBER & DIGESTER SECTION "B"

SHEET C5.3

DATE NOV 2, 2022

SCALE AS SHOWN

JOB NUMBER 2020-070

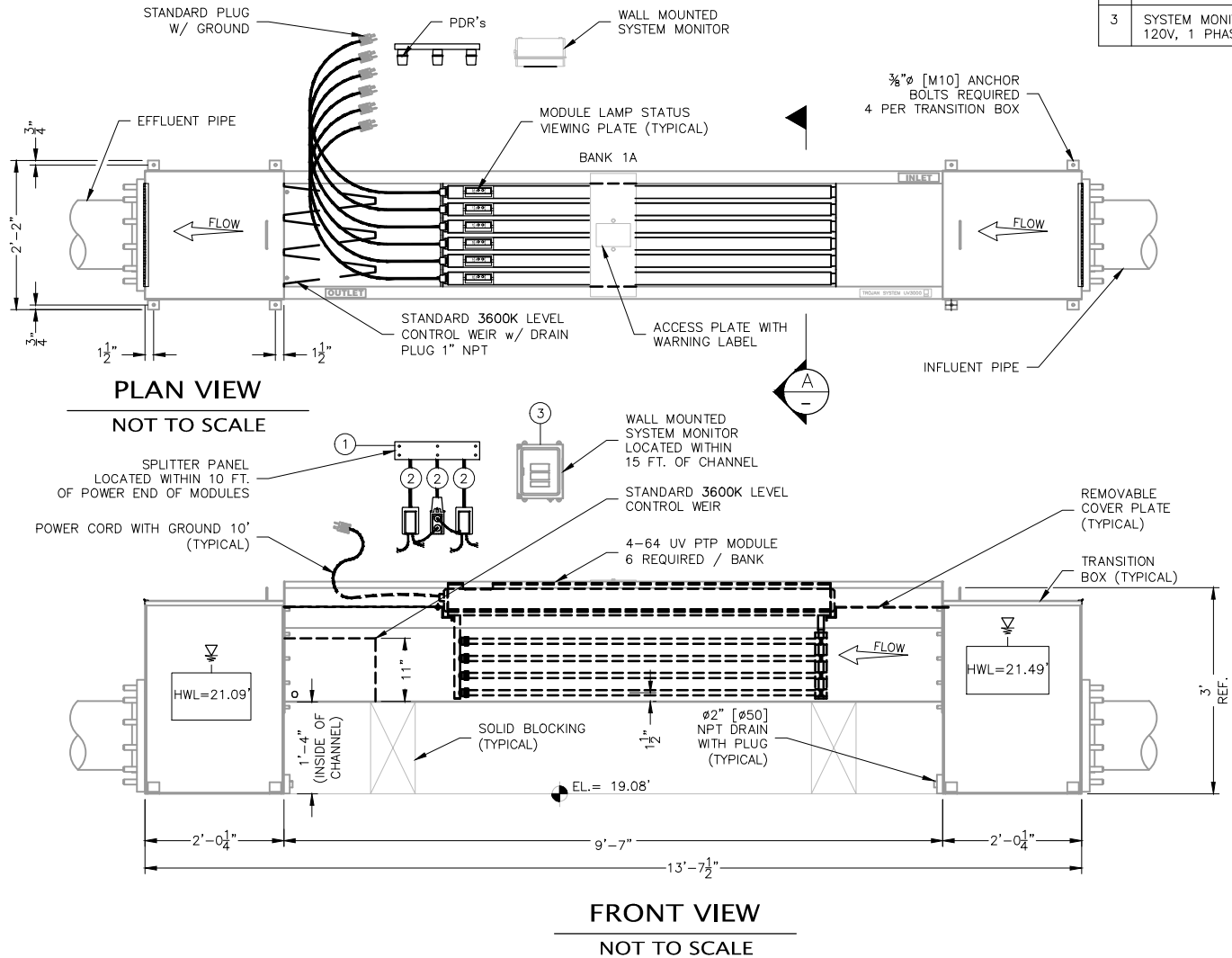
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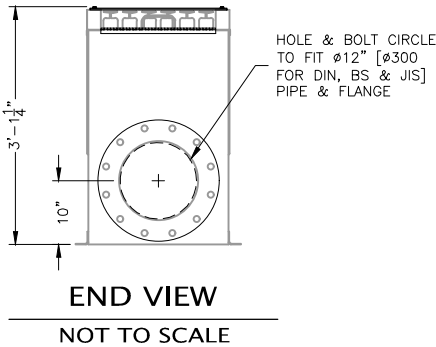
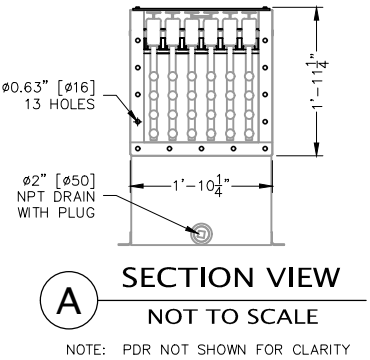
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W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 C5.1 UV DISINFECTION & DIGESTER.DWG - 11/2/2022 3:26 PM - Jeff Smith

NOTES:
: DO NOT SLOPE CHANNEL FLOOR.
: CHANNEL WIDTH & DEPTH MUST BE KEPT WITHIN A TOLERANCE OF + OR - 1/4".
: CONTRACTOR IS RESPONSIBLE FOR SUPPLYING ALL ANCHOR BOLTS.
: ELECTRICAL REQUIREMENTS SHOWN ARE TO SUPPLY TROJAN UV EQUIPMENT ONLY. ELECTRICAL INRUSH FACTOR TO BE ADDED AS PER LOCAL CODE.
: ANY EXTRA OUTLETS NOT BEING USED BY TROJAN EQUIPMENT HAVE NOT BEEN INCLUDED IN THE INTERCONNECT AMPERAGE.
: CONTRACTOR TO REVIEW ALL TROJAN TECHNOLOGIES INSTALLATION INSTRUCTIONS PRIOR TO EQUIPMENT INSTALLATION.
: DO NOT ENCASE THE STEEL CHANNEL IN CONCRETE.



EQUIPMENT INTERCONNECTIONS			
No.	DESCRIPTION	FROM	TO
1	SPLITTER PANEL POWER SUPPLY 120V, 1 PHASE, 2 WIRE, ACTUAL DRAW 19.1 AMPS / SPLITTER PANEL	DISTRIBUTION PANEL (DP) (NOT SHOWN)	SPLITTER PANEL
2	POWER DISTRIBUTION RECEPTACLE (PDR) POWER SUPPLY 120V, 1 PHASE, 2 WIRE, ACTUAL DRAW 6.3 AMPS / PDR	SPLITTER PANEL	PDR
3	SYSTEM MONITOR POWER SUPPLY 120V, 1 PHASE, 2 WIRE, 5 AMPS	DP (NOT SHOWN)	SYSTEM MONITOR



1 UV DISINFECTION UNIT
NOT TO SCALE



BID SET

NO.	REVISIONS	BY	DATE
3	ADD UV DISINFECTION	JGC	5-10-22

11-02-2022

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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

UV DISINFECTION EQUIPMENT

SHEET
C5.4

DATE
NOV 2, 2022

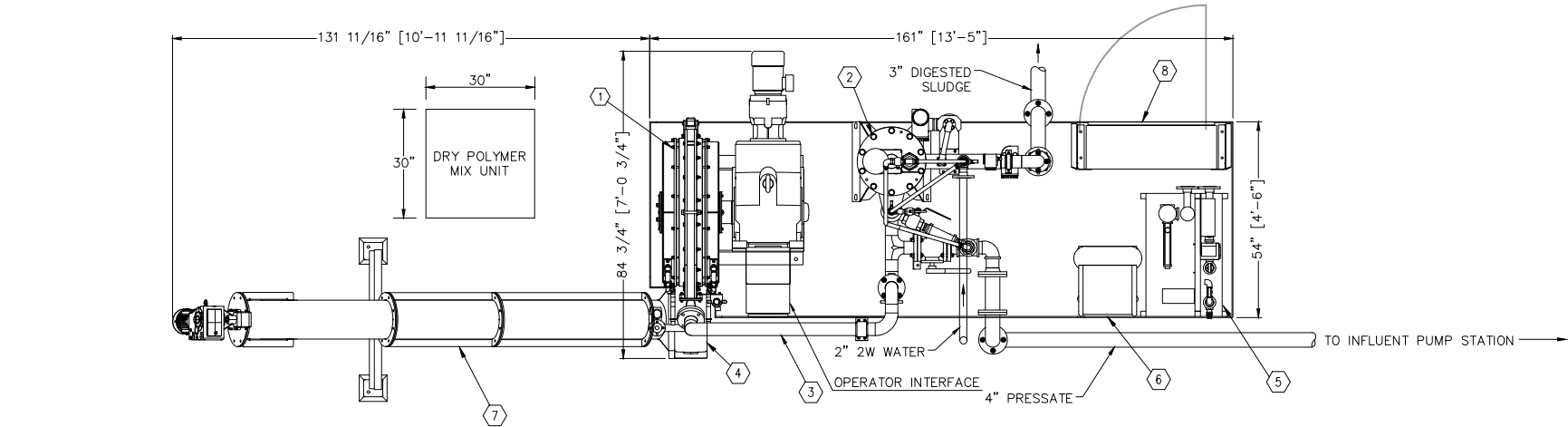
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JOB NUMBER
2020-070

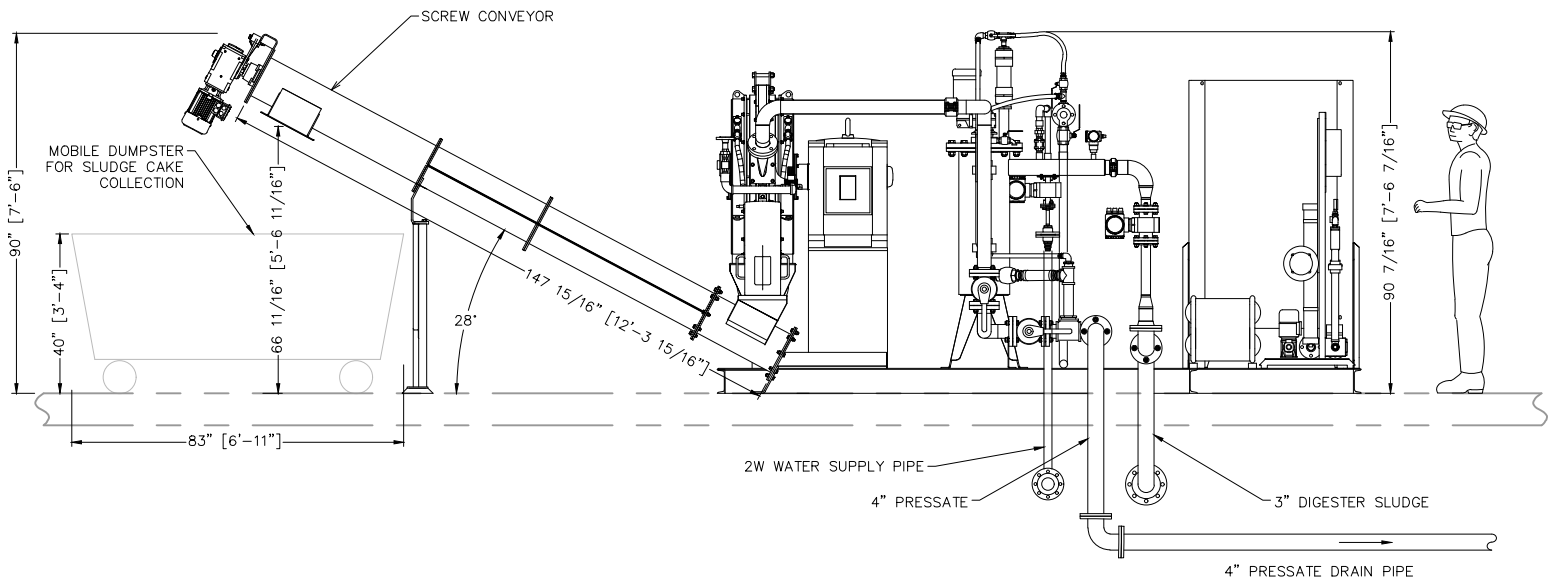
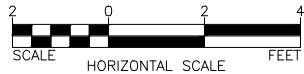
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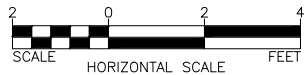
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W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020_2020-070 C6.1 ROTARY PRESS SYSTEM.DWG - 11/2/2022 3:26 PM - Jeff Smith



ROTARY FAN PRESS PLAN



ROTARY FAN PRESS ELEVATION



NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

MATERIAL LIST						
ITEM	REF.	QTY	DESIGNATION	MAT.	DIM.	NOTES
1		1	ROTARY PRESS MODEL	STEEL		SECTION 46 76 26
2		1	FLOCCULATOR	STEEL		SECTION 46 76 26
3		1	SLUDGE MANIFOLD AND PIPING	S.S.316		SECTION 46 76 26
4		1	CAKE CHUTE	LLDPE		SECTION 46 76 26
5		1	EMULSION POLYMER MAKE UP SYSTEM			SECTION 46 33 33
6		1	AIR COMPRESSOR			SECTION 46 76 26
7		1	SCREW CONVEYOR EQUIPMENT	S.S.		SECTION 46 12 13
8		1	CONTROL PANEL			SECTION 46 76 26



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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

ROTARY FAN PRESS SYSTEM

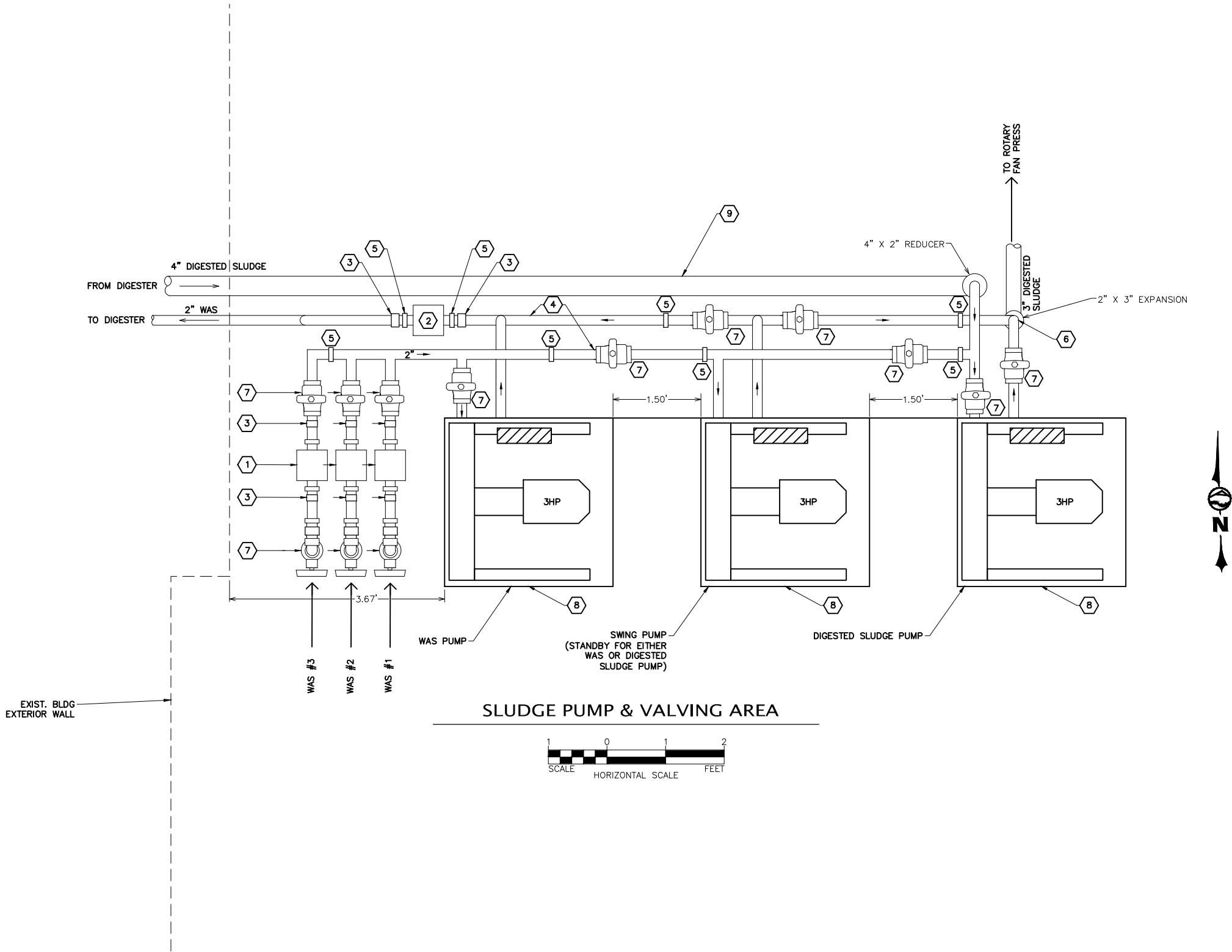
SHEET
C6.1

DATE
NOV 2, 2022

SCALE
AS SHOWN

JOB NUMBER
2020-070

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 C6.2 SLUDGE PUMPS.DWG - 11/2/2022 3:26 PM - Jeff Smith



- KEYED NOTES**
- 1 = SOLENOID VALVE, ASCO REDHAT 2 WAY SERIES 8221. NORMALLY CLOSED.
MAX WATER DIFFERENTIAL PRESSURE = 150 PSI.
SIZE = 2 INCH.
POWER= 120V COIL.
ORIFICE SIZE = 1.75 INCH.
BODY = BRASS, CATALOG #8221G013.
STAINLESS STEEL CORE TUBE, PLUGNUT, SPRINGS, AND PISTON.
ENCLOSURE = WATERTIGHT NEMA 4X.
CORROSION RESISTANT.
 - 2 = 2" FLOW METER PER ELECTRICAL DESIGN REQUIREMENTS
 - 3 = 2" UNION. DEMONSTRATE REMOVAL AND REATTACHMENT
 - 4 = PIPING TO BE SCH 80 PVC, 2" DIAMETER
 - 5 = SUPPORTS PER PIPE SUPPORT DETAIL (5/C7.2)
 - 6 = FILL ANNULAR SPACE WITH DOW CORNING 732 MULTI-PURPOSE SEALANT. JOINT SHALL BE WATER TIGHT
 - 7 = 2" PVC MANUAL BALL VALVE
 - 8 = SLUDGE PUMPS PER 22 13 36 PACKAGED SLUDGE PUMP EQUIPMENT
 - 9 = 4" SCHEDULE 80 PVC
- GENERAL NOTE**
- COORDINATE PIPING ALIGNMENT WITH OWNER. FIELD CONFIGURE FOR BEST ACCESS.



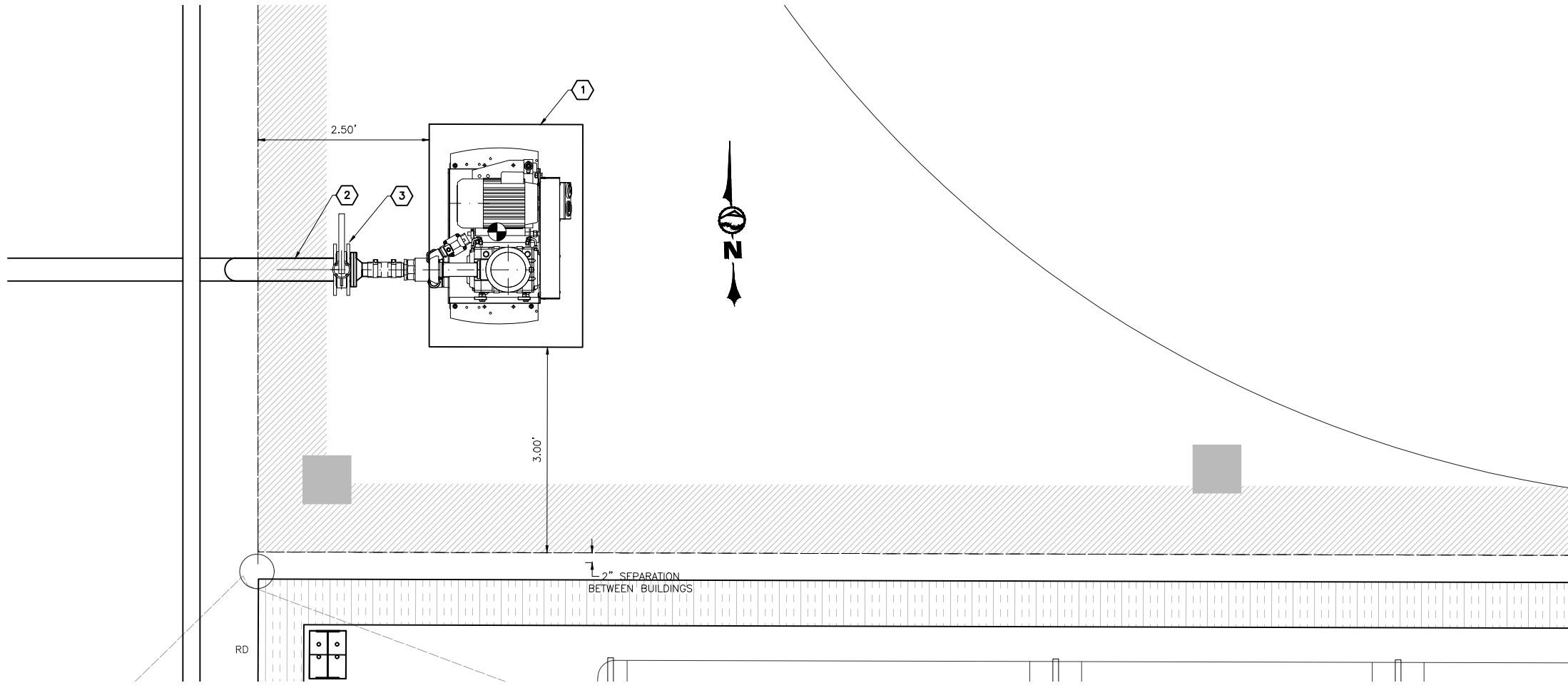
BID SET

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

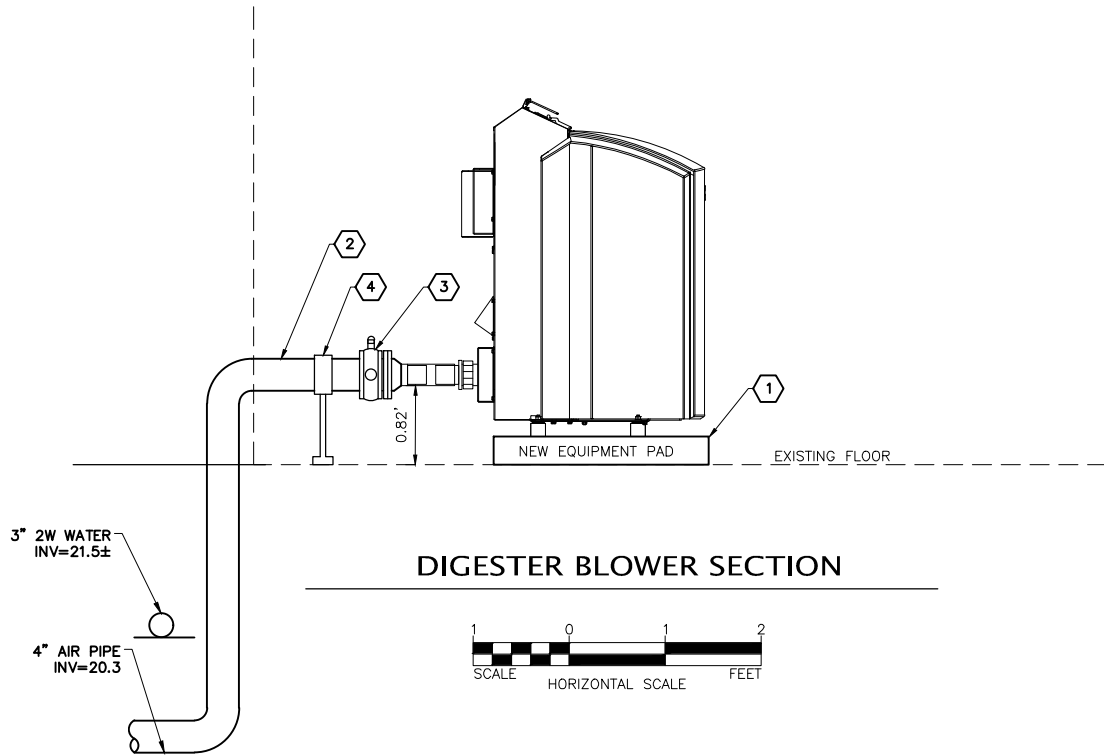
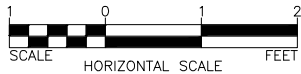
11-02-2022

WILSON ENGINEERING WILSONENGINEERING.COM			
DESIGNED BY	JGC	DRAWN BY	JGS
CHECKED BY	AWL		
EASTSOUND SEWER AND WATER DISTRICT		WASHINGTON	
SAN JUAN COUNTY		WASTEWATER TREATMENT PLANT UPGRADE - PHASE I	
DATE		NOV 2, 2022	SCALE
SHEET		C6.2	AS SHOWN
PAGE		39	JOB NUMBER
OF		96	2020-070
		SLUDGE PUMP AND VALVING AREA	

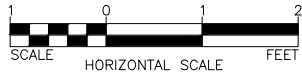
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W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 C6.3 DIGESTER BLOWER.DWG - 11/2/2022 3:26 PM - Jeff Smith



DIGESTER BLOWER



DIGESTER BLOWER SECTION



- KEYED NOTES**
- 1 = 3.25'x2.25' BLOWER EQUIPMENT PAD PER (3) C7.4
 - 2 = 4" AIR PIPING (304 SS)
 - 3 = 4" BUTTERFLY VALVE
 - 4 = PIPE SUPPORT PER (5) C7.2



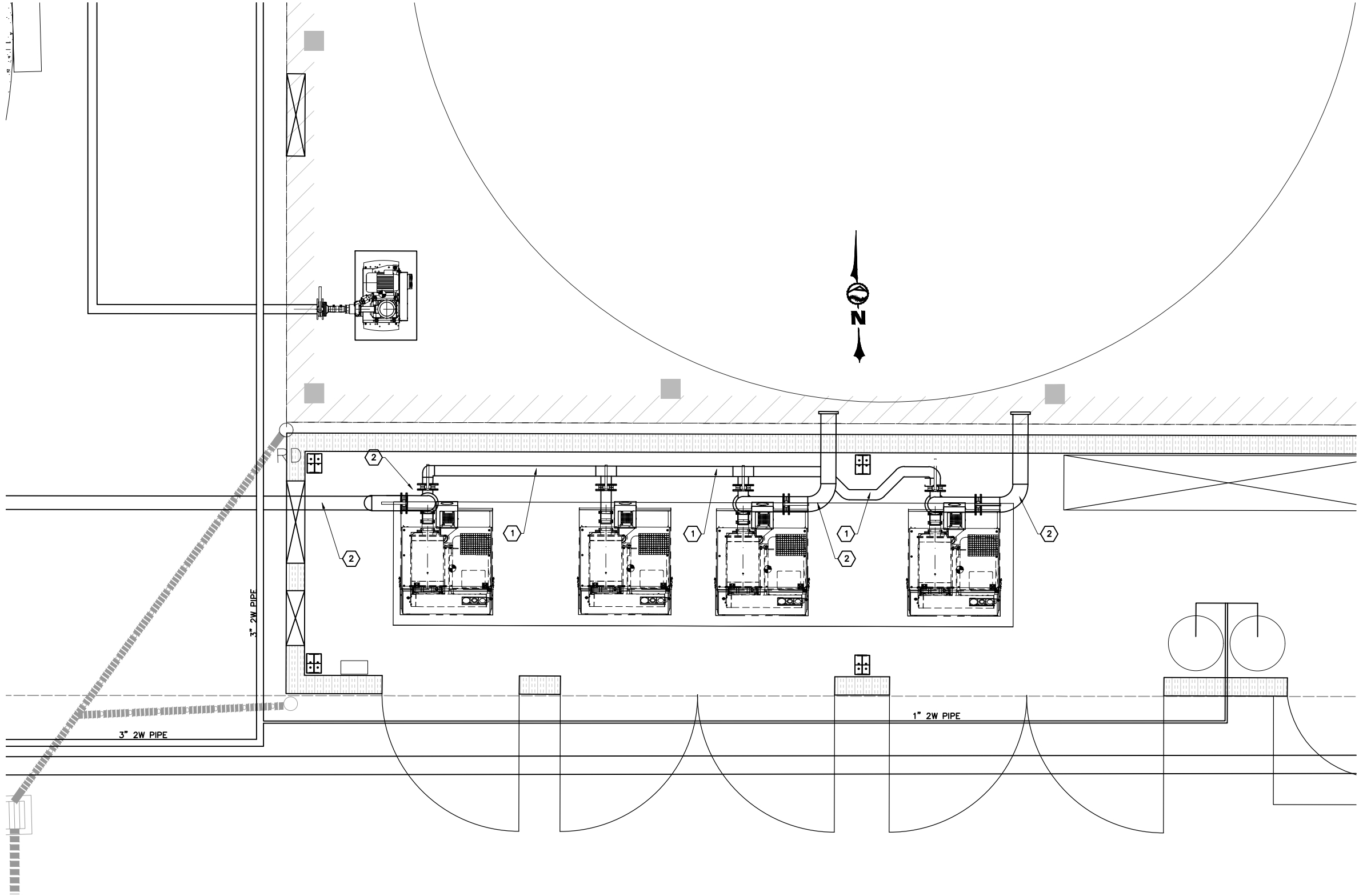
BID SET

NO.	REVISIONS	BY	DATE

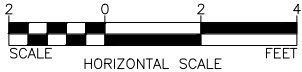
11-02-2022

SHEET C6.3	DATE NOV 2, 2022	EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY JCC			WILSON ENGINEERING	WILSONENGINEERING.COM
	SCALE AS SHOWN	SAN JUAN COUNTY WASHINGTON		DRAWN BY JGS				
PAGE 40 OF 96	JOB NUMBER 2020-070	WASTEWATER TREATMENT PLANT UPGRADE - PHASE I		CHECKED BY AWL				
		DIGESTER BLOWER						

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 Inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
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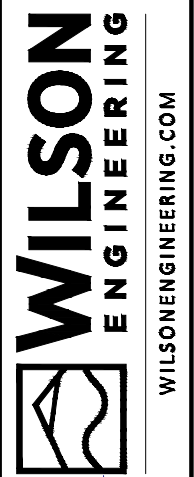


BIOLOGICAL TREATMENT BLOWERS



BID SET

- KEYED NOTES**
- 1 = 3" STAINLESS STEEL AIR PIPE. SEE EXPOSED AERATION PIPING, SECTION 33 31 00
 - 2 = 6" STAINLESS STEEL AIR PIPE. SEE EXPOSED & BURIED AERATION PIPING SECTION 33 31 00



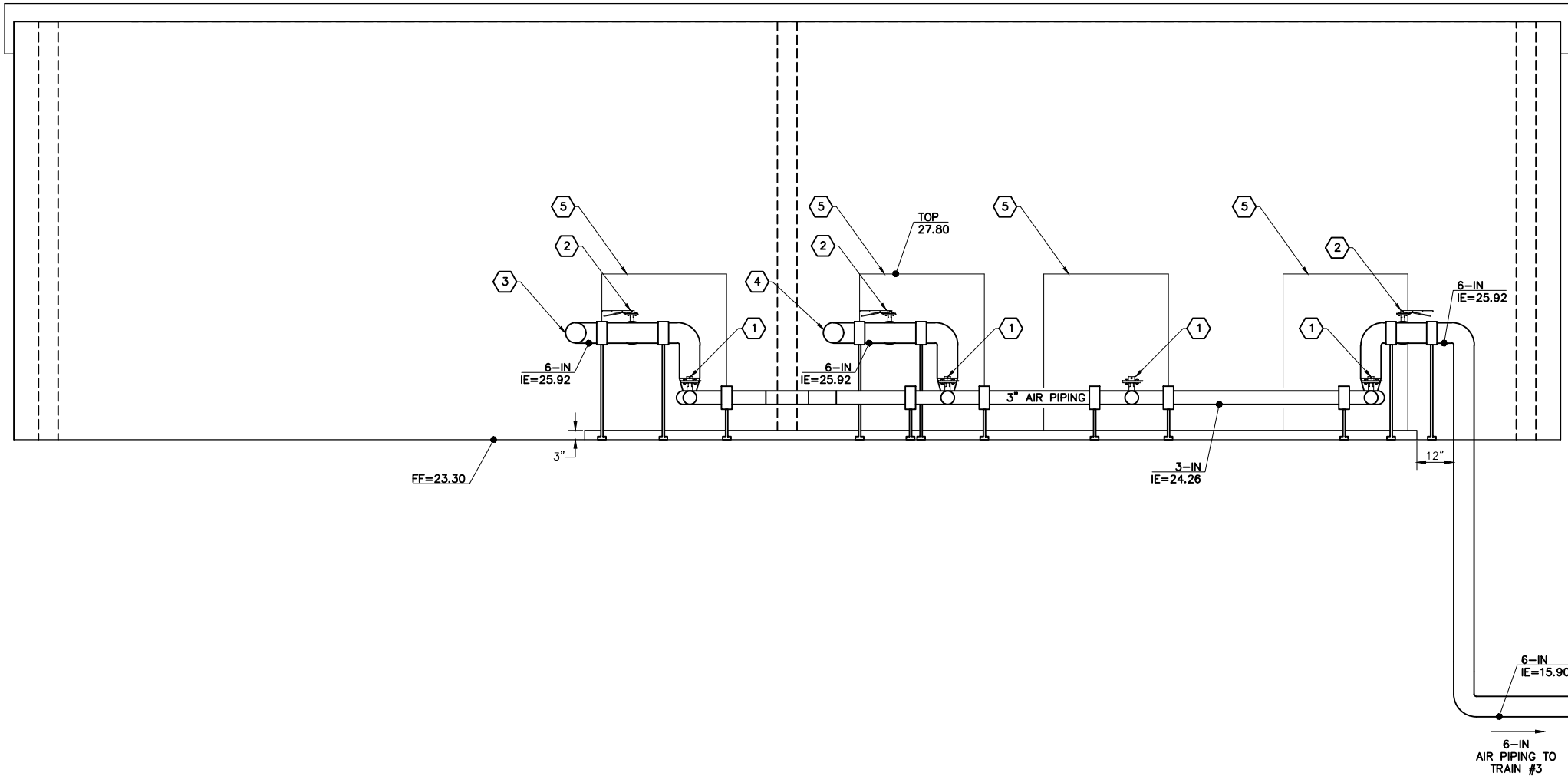
DESIGNED BY	JCC
DRAWN BY	JCS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
SAN JUAN COUNTY
BIOLOGICAL TREATMENT BLOWERS

SHEET	C6.4
DATE	NOV 2, 2022
SCALE	AS SHOWN
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PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
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BIOLOGICAL TREATMENT BLOWER ELEVATION



34
33
32
31
30
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28
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20
19
18
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16
15

- KEYED NOTES
- 1 = 3" BUTTERFLY VALVE
 - 2 = 6" BUTTERFLY VALVE
 - 3 = 6" AIR PIPING TO TRAIN #1 (TERMINATE WITH BLIND FLANGE INSIDE EXISTING METAL BUILDING)
 - 4 = 6" AIR PIPING TO TRAIN #2 (TERMINATE WITH BLIND FLANGE INSIDE EXISTING METAL BUILDING)
 - 5 = NEW BLOWER PER SPECS

11-02-2022

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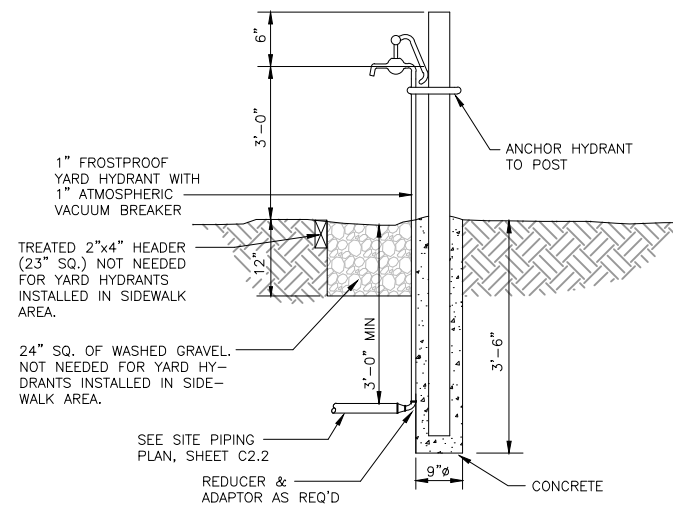


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EASTSOUND SEWER AND WATER DISTRICT					
SAN JUAN COUNTY WASHINGTON					
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I					
BIOLOGICAL TREATMENT BLOWER ELEVATION					

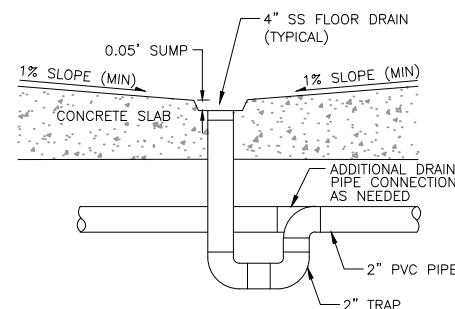
SHEET	C6.5	DATE	NOV 2, 2022	SCALE	AS SHOWN	JOB NUMBER	2020-070
PAGE	42	OF	96				



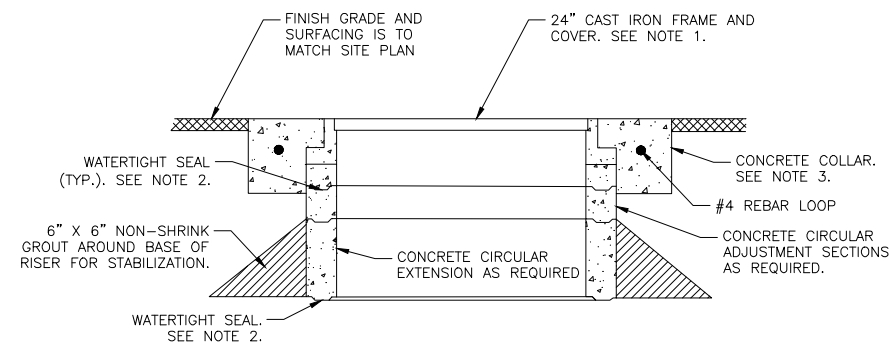
BID SET



1 TYPICAL YARD HYDRANT
NOT TO SCALE



2 TYPICAL FLOOR DRAIN
NOT TO SCALE



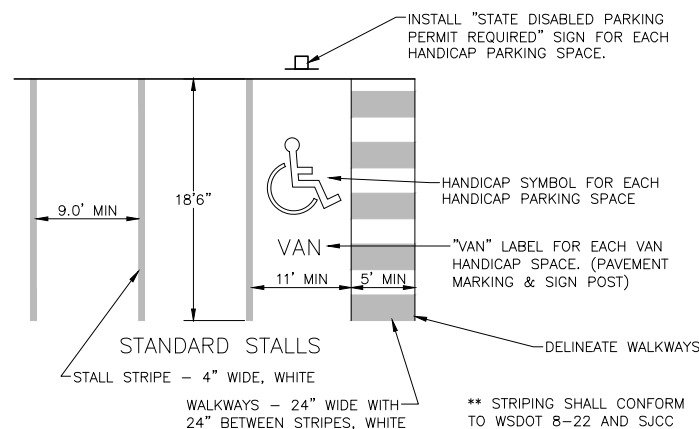
NOTES:

1. MANHOLE FRAME AND COVER SHALL BE PLACED AT GRADE AND LEVEL WITH EXISTING ROADWAY.
2. ASPHALTIC SEALANT SHALL BE USED, RAMNEK OR EQUAL, TO PROVIDE WATERTIGHT SEAL BETWEEN TANK AND RISER AND GRADE RINGS.
3. 12" WIDE BY 8" THICK STEEL REINFORCED CONCRETE COLLAR INSTALLED AROUND RISERS IN ALL TRAFFIC RATED AREAS.

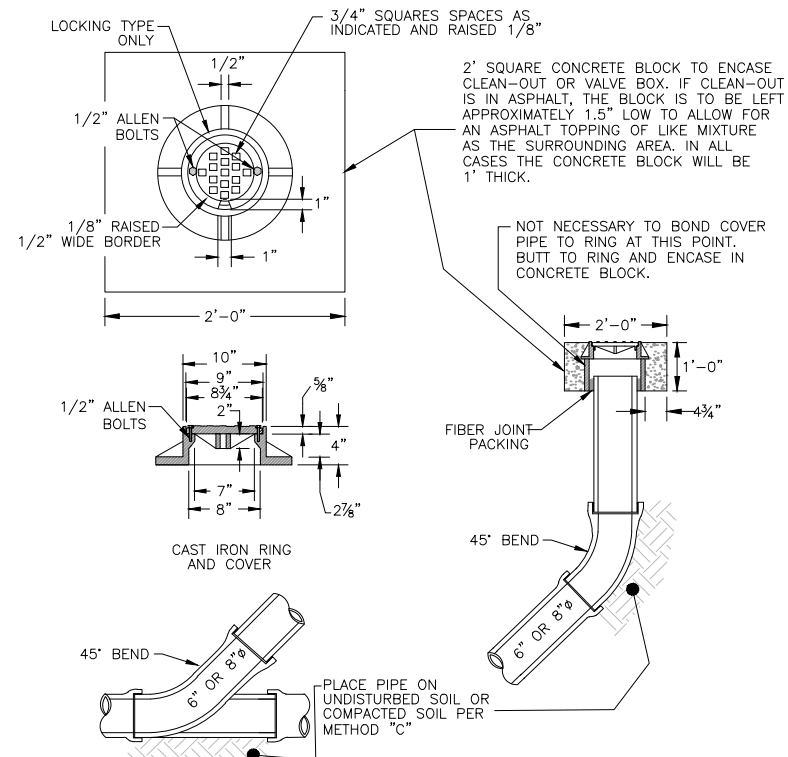
3 TRAFFIC RATED RISER AND LID DETAIL
NOT TO SCALE



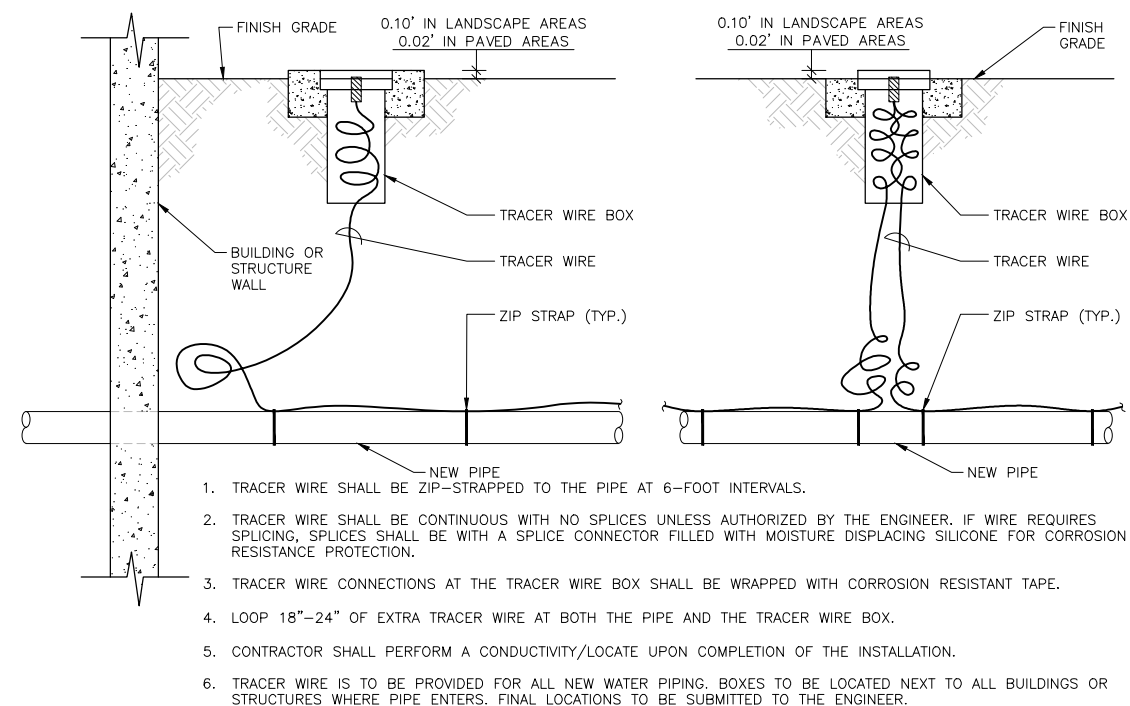
BID SET



4 PARKING STALL STRIPING

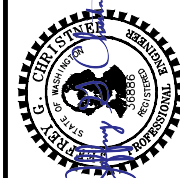


5 GRAVITY SEWER CLEANOUT DETAIL
NOT TO SCALE



6 TRACER WIRE LOCATER BOX DETAIL
NOT TO SCALE

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



DESIGNED BY	JGC
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CHECKED BY	AWL

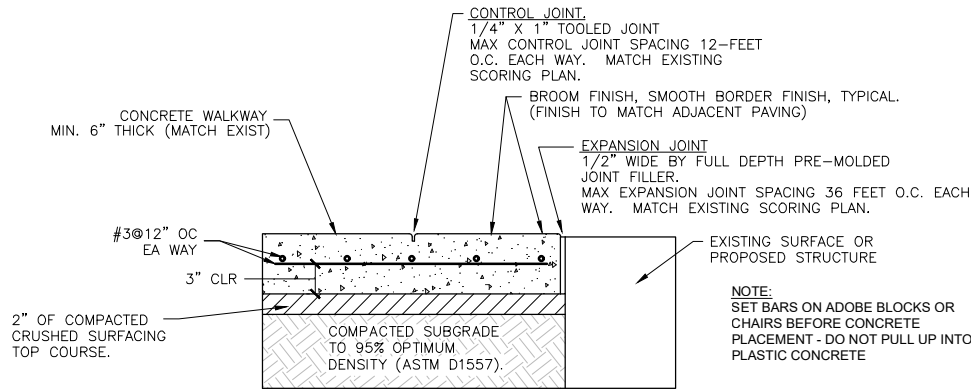
EASTSOUND SEWER AND WATER DISTRICT

WASCO COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

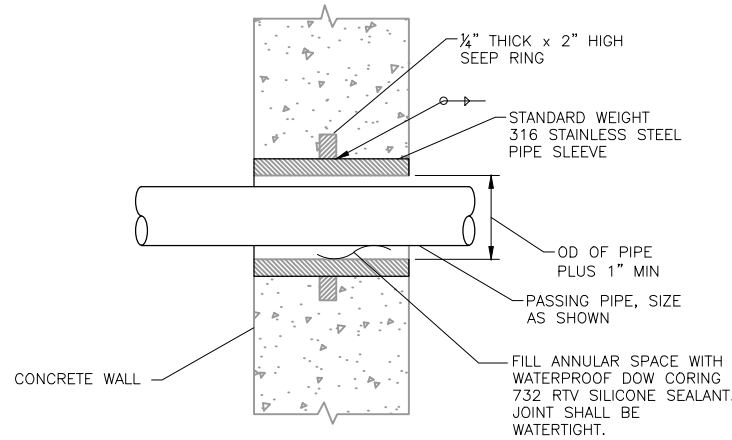
CIVIL DETAILS

SHEET C7.1	PAGE	43 OF 96
	DATE	NOV 2, 2022
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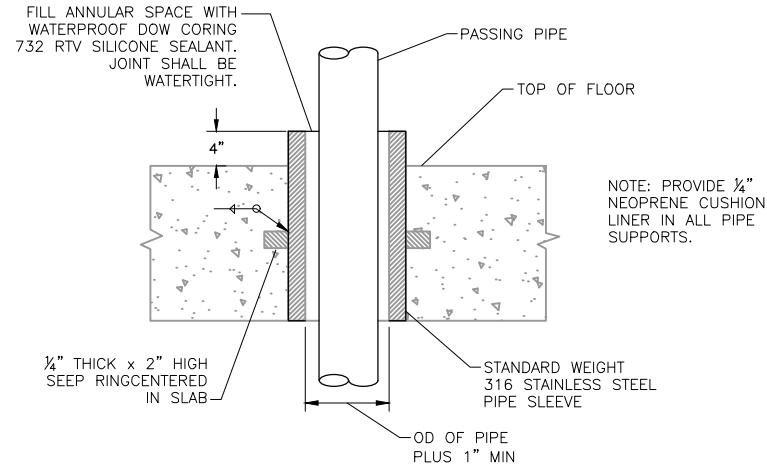
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1 CONCRETE WALKWAY DETAIL
NOT TO SCALE



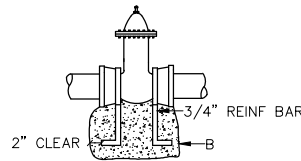
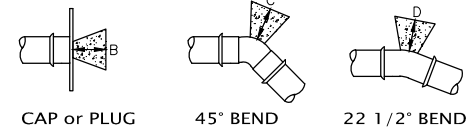
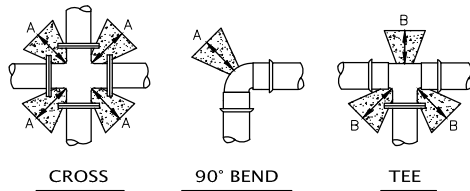
2 PIPE SLEEVE DETAIL
NOT TO SCALE



3 FLOOR SLEEVE DETAIL
NOT TO SCALE

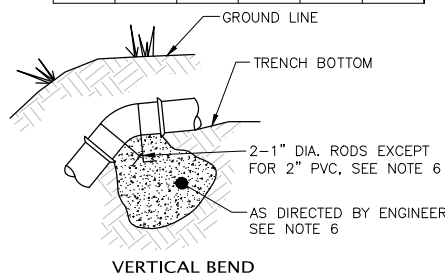


BID SET



(PARTIAL RESTRAINT MUST BE PROVIDED BY PIPELINE BEYOND VALVE)

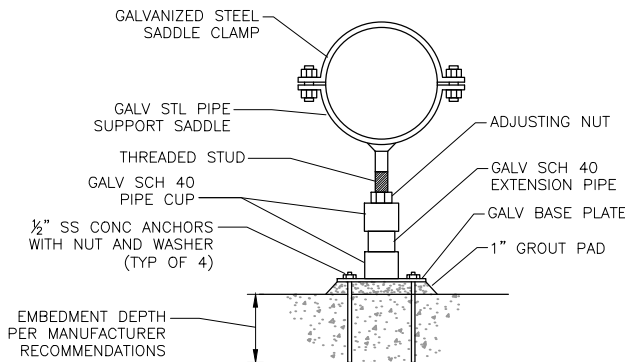
THRUST BLOCK TABLE					
MINIMUM BEARING AREA AGAINST UNDISTURBED SOIL IN SQUARE FEET					
PIPE SIZE	A	B	C	D	E
4"	2	2	2	2	2
6"	4	3	2	2	2
8"	7	5	4	2	2
10"	11	8	6	3	2
12"	16	12	9	5	3
16"	29	20	16	8	4
20"	45	32	24	13	6



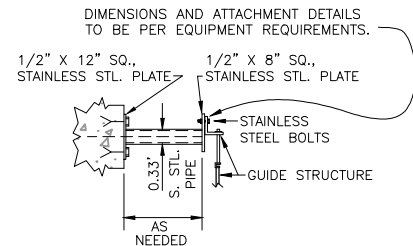
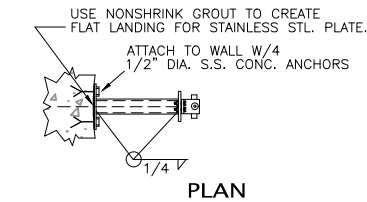
- NOTES:
1. SQUARE FEET OF CONCRETE THRUST BLOCK AREA IS BASED ON 200 P.S.I. INTERNAL PRESSURE, A SOIL SAFE BEARING OF 3000 POUNDS PER SQUARE FOOT AND A FACTOR OF SAFETY OF 1.5.
 2. BEARING AREA MUST BE ADJUSTED FOR INTERNAL PRESSURES AND LOWER SOIL BEARING VALUES.
 3. CONCRETE BLOCKING SHALL BE CAST IN PLACE AND HAVE A MINIMUM OF 1/4 SQUARE FOOT BEARING AGAINST THE FITTING.
 4. BLOCK SHALL BEAR AGAINST FITTINGS ONLY AND SHALL BE CLEAR OF JOINTS TO PERMIT TAKING UP OR DISMANTLING JOINT.

5. THE CONTRACTOR SHALL INSTALL BLOCKING WHICH IS ADEQUATE TO WITHSTAND FULL TEST PRESSURE AS WELL AS TO CONTINUOUSLY WITHSTAND OPERATING PRESSURE UNDER ALL CONDITIONS OF SERVICE.
6. STAINLESS STEEL BANDING SHALL BE USED AT 2" PVC VERTICAL BENDS INSTEAD OF 1" RODS. CONTACT ENGINEER FOR SIZING OF THRUST BLOCK AND DETAILS.
7. ALL BENDS, TEES & CROSSES SHALL INCLUDE RESTRAINED JOINTS (ROMAC GRIPPER) AS WELL AS THRUST BLOCKING.

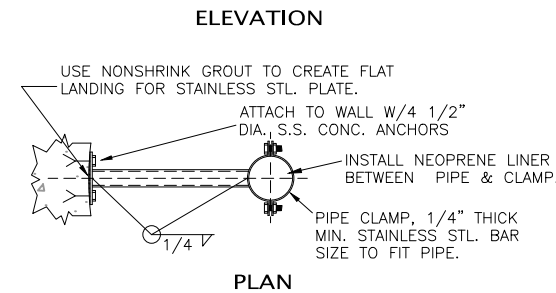
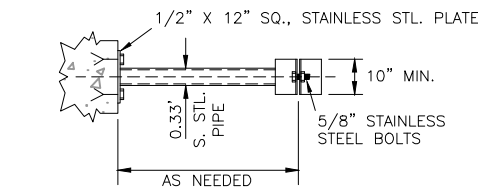
4 FORCEMAIN THRUST BLOCKING SCHEDULE
NOT TO SCALE



5 PIPE SUPPORT DETAIL
NOT TO SCALE

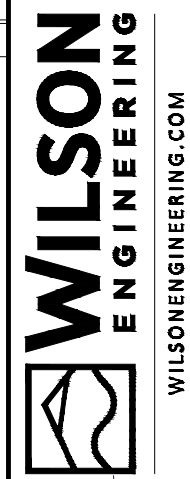


6 GUIDE STRUCTURE SUPPORT
NOT TO SCALE



7 VERTICAL PIPE BRACE
NOT TO SCALE

11-02-2022



DESIGNED BY JCC
DRAWN BY SNR
CHECKED BY AWL

EASTSOUND SEWER AND WATER DISTRICT

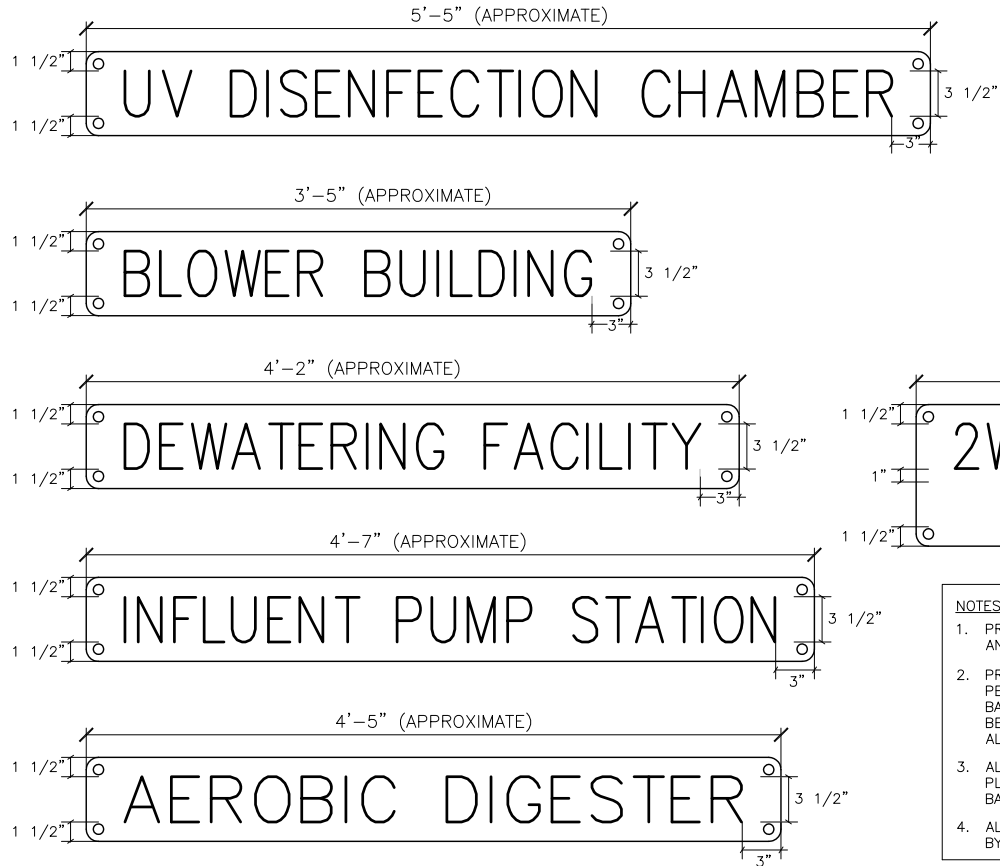
WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

CIVIL DETAILS

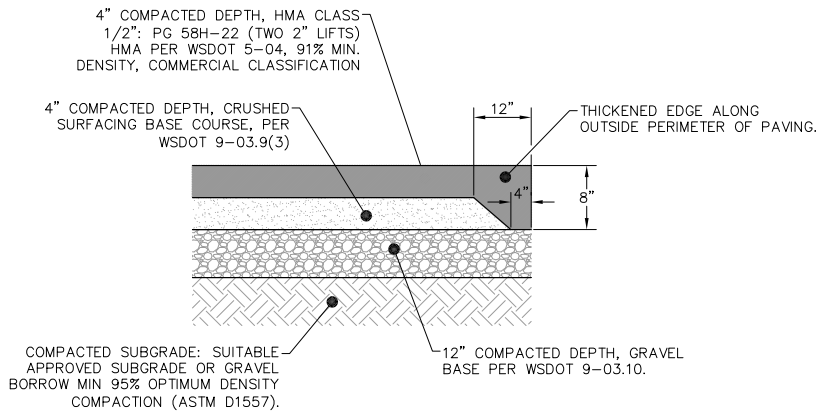
DATE NOV 2, 2022
SCALE AS SHOWN
JOB NUMBER 2020-070

SHEET C7.2
PAGE 44 OF 96

PLOT SETTINGS: Adobe PDF.pc3, Ledger, Portrait, 1:2, WE APWA_UNSCREENED.ctb
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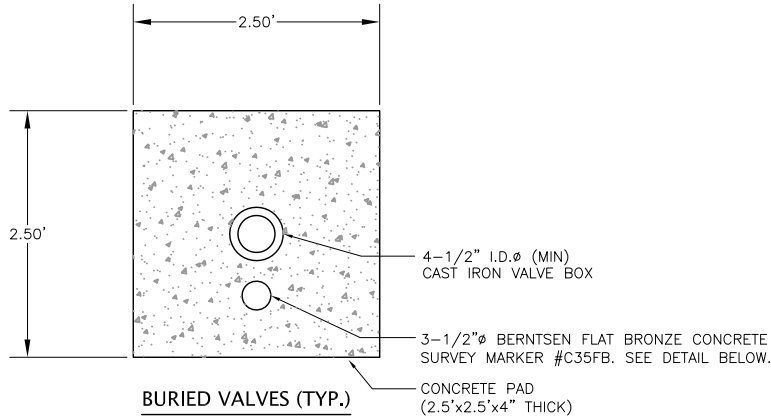


1 WWTP TANK SIGN DETAILS
NOT TO SCALE



3 ASPHALT CONCRETE PAVEMENT WITH THICKENED EDGE
NOT TO SCALE

- NOTES FOR SIGNS:
1. PROVIDE FOUR STAINLESS STEEL CONCRETE ANCHORS PER WALL SIGN.
 2. PROVIDE FOUR 2" S.S. GUARD RAIL FASTENERS PER RAIL SIGN. PROVIDE & ATTACH ALUMINUM BACKING PLATE TO FASTENERS. CENTER BASIN SIGN BETWEEN HORIZONTAL RAILS AND MOUNT TO ALUMINUM BACKING PLATE.
 3. ALL SIGNS ARE TO BE BAKED ENAMEL ON STEEL PLATE WITH BLACK LETTERING ON A WHITE BACKGROUND.
 4. ALL SIGN LOCATIONS ARE TO BE FIELD DETERMINED BY OWNER.



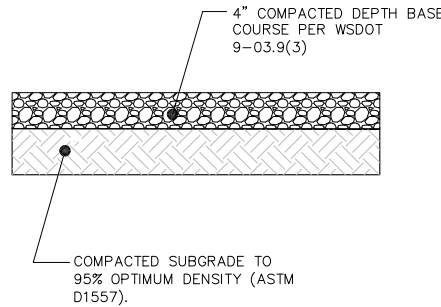
1/4" TALL BERNTSEN STANDARD TEXT STYLE TO BE STAMPED BY BERNTSEN. THE TEXT "VALVE #" IS TO BE OFFSET AS SHOWN TO ALLOW ROOM FOR THE ENGRAVED ONE OR TWO DIGIT VALVE NUMBER PRECEDING IT TO BE CENTERED WITHIN THIS LINE OF TEXT. THIS TEXT TO REMAIN THE SAME PER EACH VALVE LABEL MARKER.

- NOTES:
- 1) 14 BRONZE VALVE LABELING MARKERS TO BE FURNISHED AND INSTALLED AT LOCATIONS ACCORDING TO THE VALVE LABELING SCHEDULE.
 - 2) SHEET C2.4 & C2.5 IS SCHEMATIC AND IS NOT INTENDED TO SHOW PRECISE LOCATIONS OF THE VALVES TO BE MARKED.

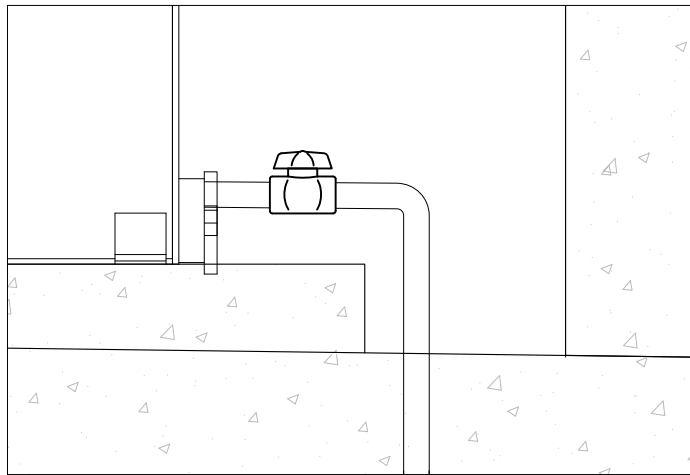
4 TYPICAL BURIED VALVE LABEL MARKER
NOT TO SCALE

BURIED VALVE IDENTIFICATION LEGEND	
1	6" GATE VALVE FOR INFLUENT FLOW METER ISOLATION
2	8" GATE VALVE FOR TRAIN 2 FLOW METER ISOLATION
3	8" GATE VALVE FOR TRAIN 3 FLOW METER ISOLATION
4	4" GATE VALVE FOR DRAIN FLOW METER ISOLATION
5	4" GATE VALVE FOR DRAIN FLOW METER ISOLATION
6	4" GATE VALVE FOR CCB OUTLET BOX DRAIN
7	4" GATE VALVE FOR CCB1 DRAIN
8	4" GATE VALVE FOR CCB 1 DRAIN
9	4" GATE VALVE FOR CCB 1 DRAIN
10	4" GATE VALVE FOR CCB 1 DRAIN
11	4" GATE VALVE FOR CCB 1 DRAIN
12	4" GATE VALVE FOR CCB 2 DRAIN
13	4" GATE VALVE FOR CCB 2 DRAIN
14	4" GATE VALVE FOR CCB 2 DRAIN
15	4" GATE VALVE FOR CCB 2 DRAIN
16	4" GATE VALVE FOR CCB 2 DRAIN
17	8" GATE VALVE FOR EFFLUENT FLOW METER ISOLATION

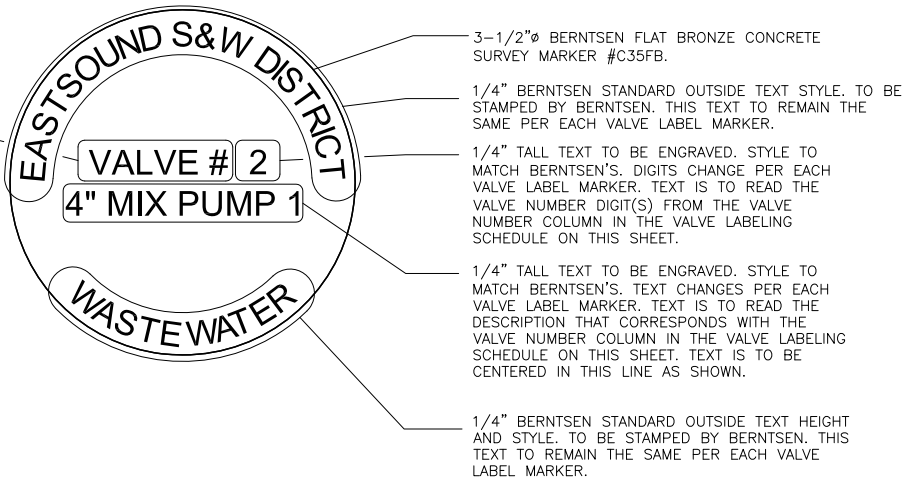
2 BURIED VALVE LABELING SCHEDULE
NOT TO SCALE



5 GRAVEL SURFACING SECTION
NOT TO SCALE



6 1" PVC BALL VALVE
NOT TO SCALE



BID SET

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

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DESIGNED BY: JGC
DRAWN BY: SNR
CHECKED BY: AWL

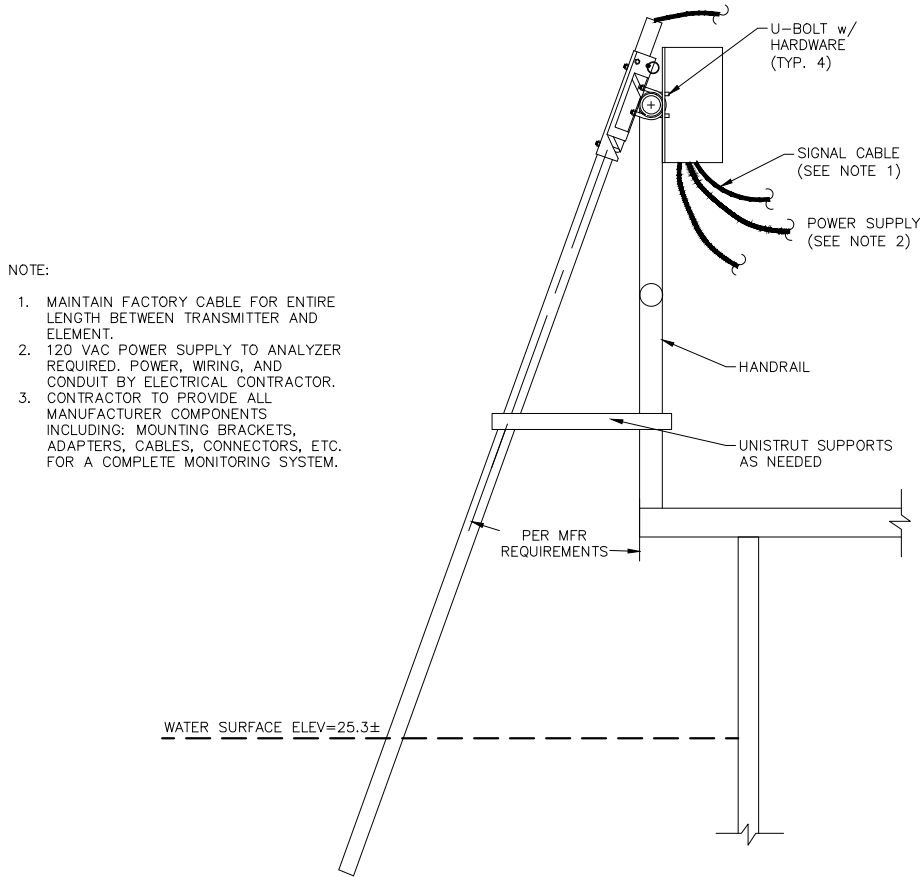
EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
CIVIL DETAILS

DATE: NOV 2, 2022
SCALE: AS SHOWN
JOB NUMBER: 2020-070

SHEET: C7.3
PAGE: 45 OF 96

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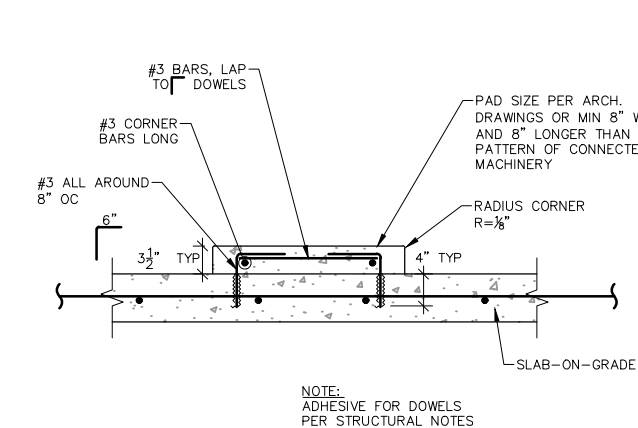
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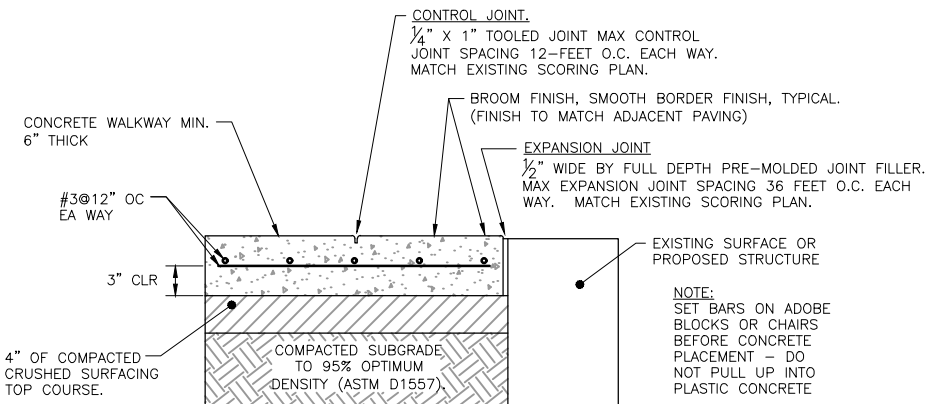
1 D.O. PROBE ELEVATION (TYP.)
NOT TO SCALE



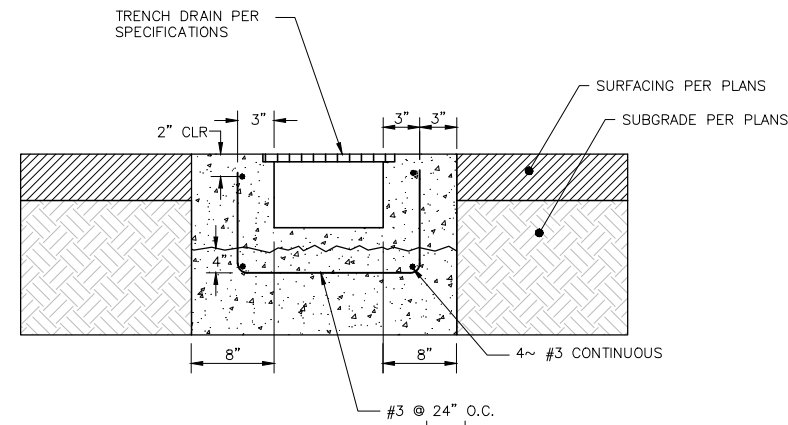
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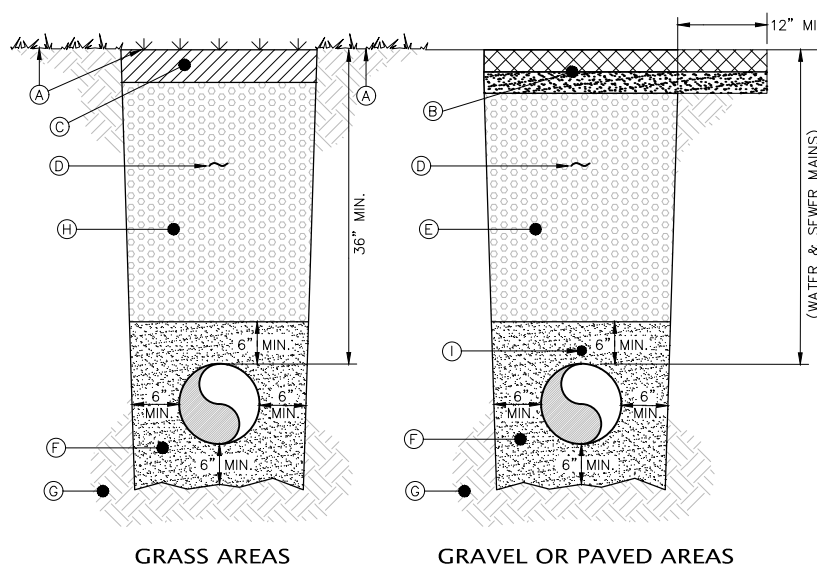
3 TYPICAL MECHANICAL EQUIPMENT PAD
NOT TO SCALE



4 CONCRETE PAVEMENT DETAIL
NOT TO SCALE



5 TRENCH DRAIN DETAIL
NOT TO SCALE



2 TYPICAL TRENCHING & BACKFILL
NOT TO SCALE

NO.	REVISIONS	BY	DATE

KEYED NOTES

- HYDROSEED EXPOSED AREAS.
- GRAVEL: NEW CRUSHED SURFACING TOP COURSE (CSTC) LAYER PER WSDOT 9-03.9(3). PAVED: HMA CL 1/2" APPLIED IN 2 LIFTS MIN. TO THICKNESS MATCHING EX. SECTION (4" MIN.) PER WSDOT 5-04.2, WITH 2" CSTC LAYER BELOW.
- NEW TOPSOIL LAYER MIN 4".
- 2" METALLIC DETECTOR TAPE 8" TO 12" BELOW FINISH GRADE.
- BANK RUN GRAVEL BACKFILL PER WSDOT 9-03.19 COMPACTED TO 95% MAX. DENSITY INSIDE RIGHT-OF-WAY.
- PIPE ZONE GRAVEL BEDDING PER WSDOT 9-03.12(3) COMPACTED TO 95% MAX. DENSITY.
- UNDISTURBED NATIVE MATERIAL.
- SELECT NATIVE BACKFILL COMPACTED TO 95% MAX. DENSITY.
- #10 AWG INSULATED TRACER WIRE ON NONMETALLIC PIPE (IN ADDITION TO 2" DETECTOR TAPE)

11-02-2022

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EASTSOUND SEWER AND WATER DISTRICT					
WASHINGTON					
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I					
SAN JUAN COUNTY					
CIVIL DETAILS					
DATE NOV 2, 2022		SCALE AS SHOWN		JOB NUMBER 2020-070	
SHEET C7.4		PAGE 46		OF 96	

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811 Call 811
two business days
before you dig

BID SET

CONTRACTOR TO DEMONSTRATE
REMOVAL AND REPLACEMENT OF
HYDRANT COMPONENTS WITHOUT
CONFLICT WITH POST AND HOSE RACK.

HEAVY DUTY ALUMINUM HOSE RACK FASTENED TO POST (RACK NOT TO INTERFERE WITH
ACCESS TO AND OPERATION OF HYDRANT).

2" SQUARE HOT DIPPED GALVANIZED STEEL SIGN POSTS.

WARNING SIGN
SEE DETAIL

DRILL FOR 3/8" DIA
FASTENERS, TYP.

3/16" THICK

3 1/2"

3/4" R CORNERS, TYP.

WALL MOUNTED SIGN & RACK

CAUTION:
2W PLANT WATER
NON POTABLE

WARNING SIGN DETAIL

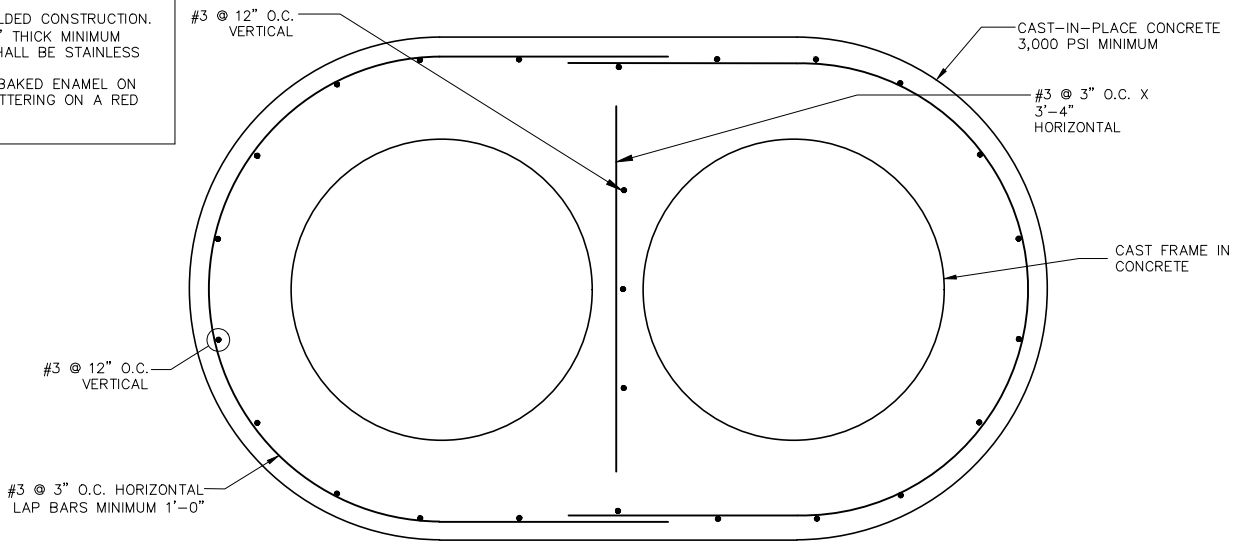
NOTES FOR YARD HYDRANT STATIONS

1. PROVIDE FREE STANDING OR WALL MOUNTED AS
APPROPRIATE.
2. HOSE RACK IS TO BE WELDED CONSTRUCTION.
MATERIAL IS TO BE 3/16" THICK MINIMUM
ALUMINUM. FASTENERS SHALL BE STAINLESS
STEEL.
3. WARNING SIGN IS TO BE BAKED ENAMEL ON
ALUMINUM WITH WHITE LETTERING ON A RED
BACKGROUND

LIGHTWEIGHT COMPOSITE MANHOLE, MORRISON BROS. CO. ID
#418LC-3800 AM, OR EQUAL, WITH STAINLESS STEEL HANDLE
AND PRY PLATE. ADDITIONAL REQUIREMENTS:

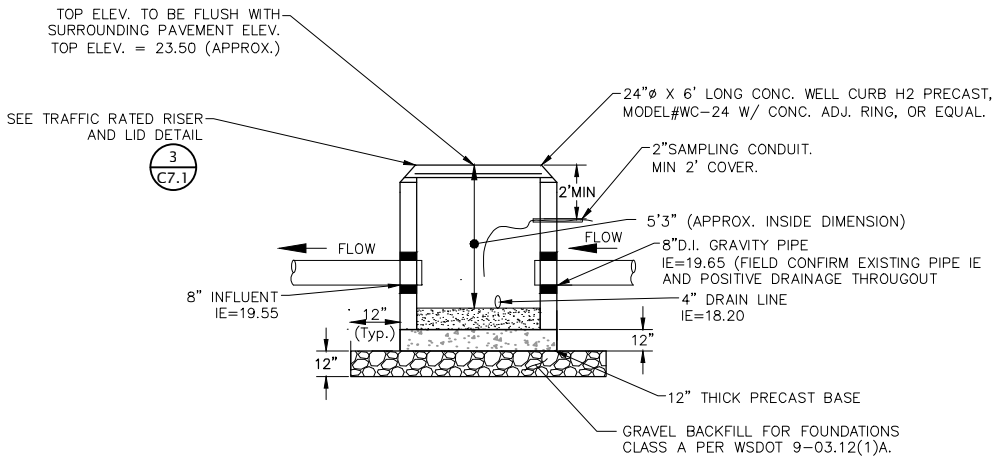
- 12 GAUGE SKIRT RING TO BE 3/16" STEEL AND 10" LENGTH
- MINIMUM ID = 35-5/16"
- LOCKING MECHANISM WITH SS CAM ARM
- GASKET LID
- COLOR = BLACK
- COVER WEIGHT = 55 LBS

DOUBLE MANHOLE CONDITION ELEVATION



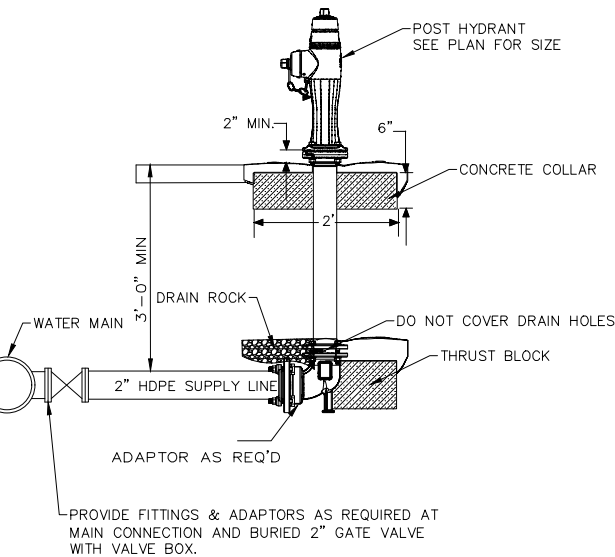
DOUBLE MANHOLE CONDITION PLAN SECTION

3 TRAFFIC MANHOLE RING DETAILS NOT TO SCALE



4 INFLUENT SAMPLE SUMP NOT TO SCALE

1 TYPICAL PLANT WATER (2W) YARD HYDRANT STATIONS NOT TO SCALE



2 POST HYDRANT DETAIL NOT TO SCALE

WILSON
ENGINEERING

WILSONENGINEERING.COM



11-02-2022

DESIGNED BY JCC
DRAWN BY SNR
CHECKED BY AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

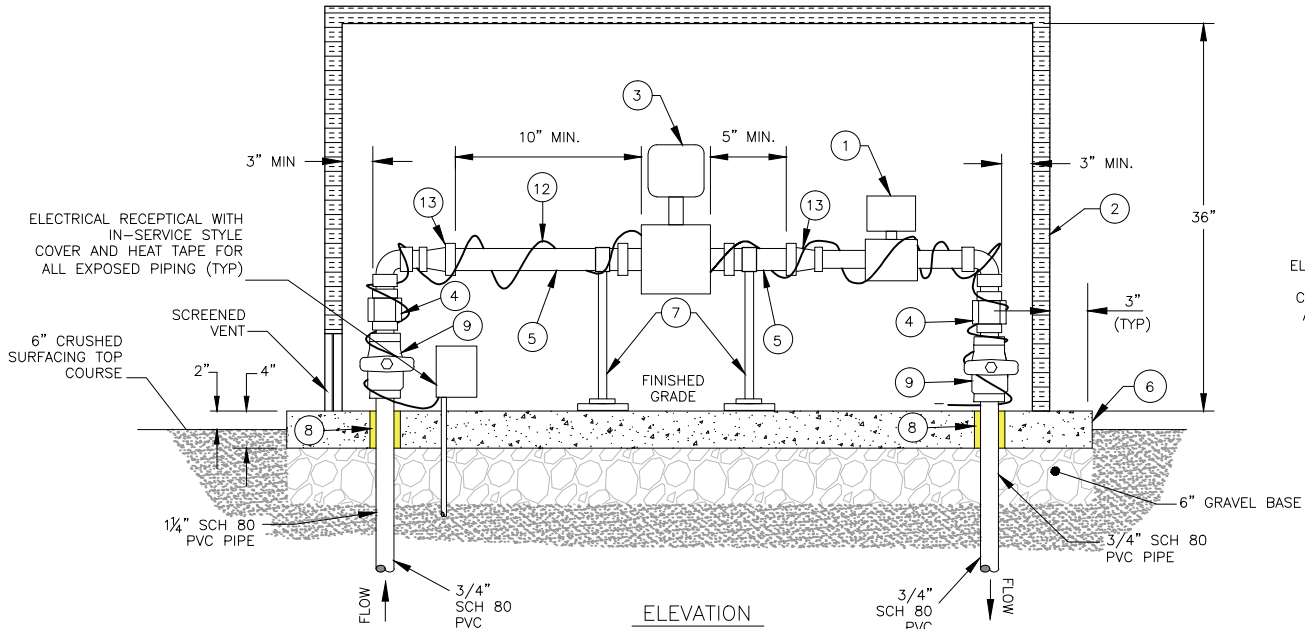
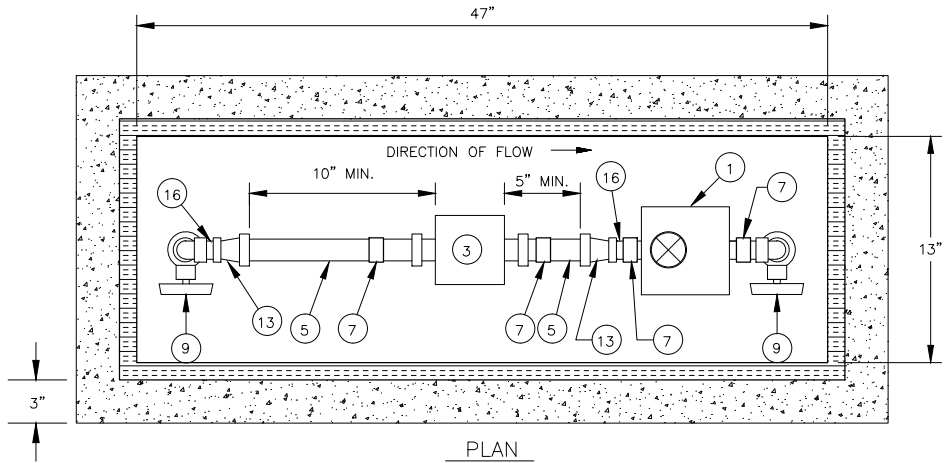
CIVIL DETAILS

DATE NOV 2, 2022
SCALE C7.5
SHEET 47 OF 96
JOB NUMBER 2020-070

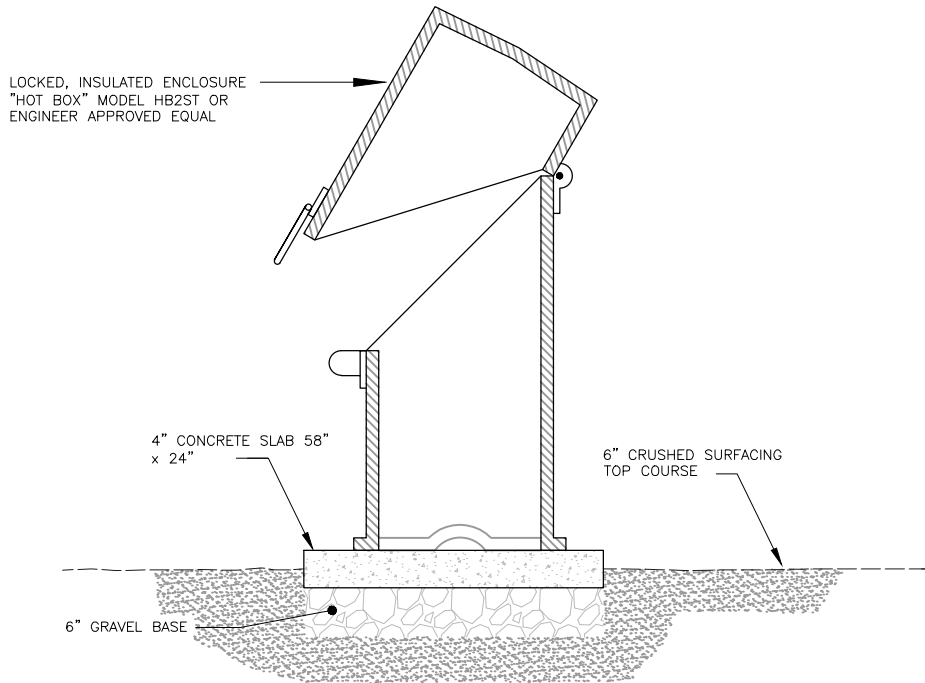
PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
W:\2020-070 EASTSOUND WWP DESIGN\DWG\2020-070 C7.1-C7.8 CIVIL DETAILS.DWG - 11/2/2022 3:26 PM - Jeff Smith

MATERIAL LIST

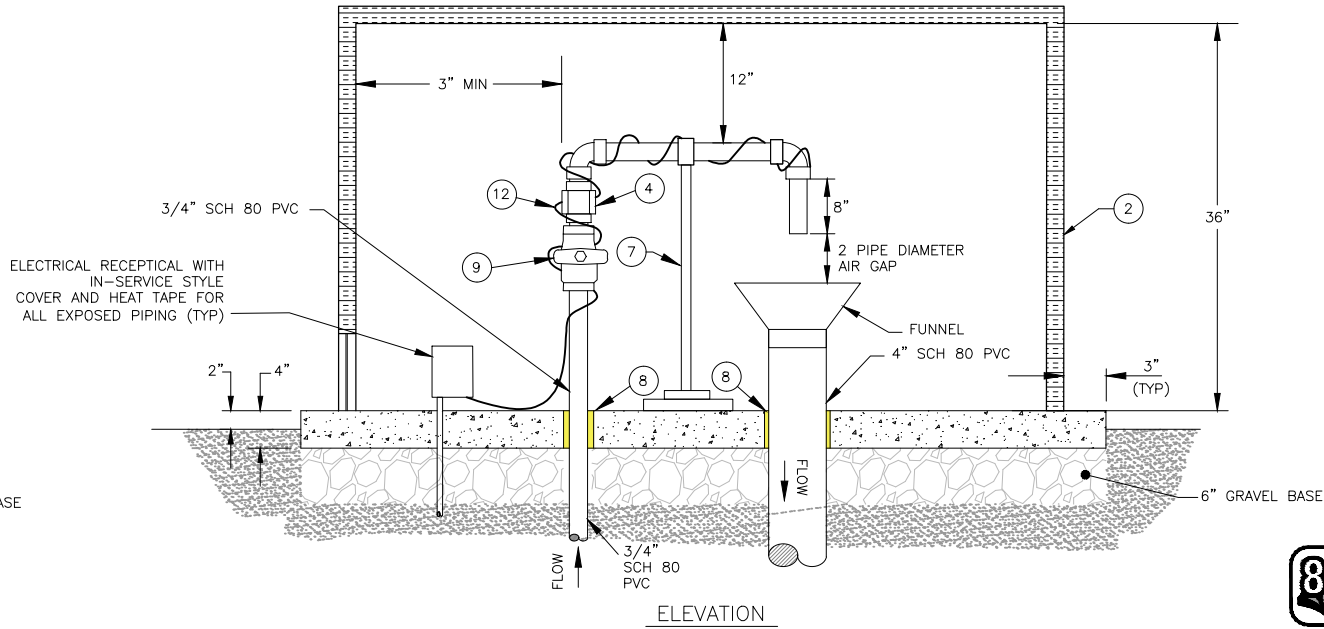
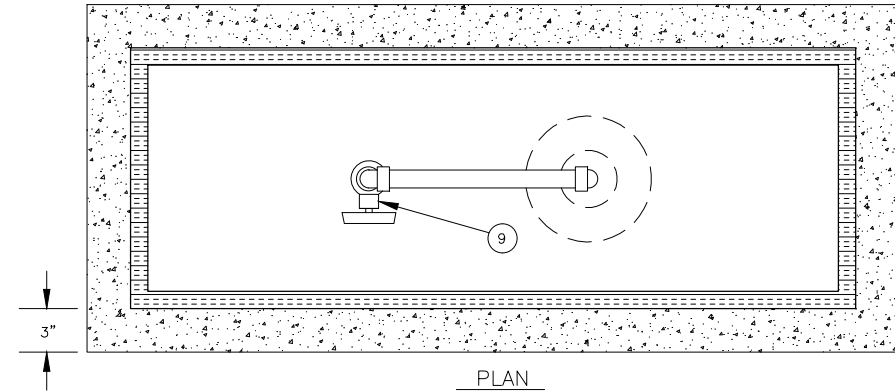
- SOLENOID VALVE, ASCO REDHAT 2 WAY SERIES 8221. SLOW CLOSING SOLENOID VALVE TO PROTECT AGAINST WATER HAMMER. NORMALLY CLOSED.
MAX WATER DIFFERENTIAL PRESSURE = 150 PSI.
SIZE = 3/4 INCH.
POWER= 120V AC, 6.1 WATTS
ORIFICE SIZE = 3/4 INCH.
BODY = BRASS, CATALOG #8221G005.
STAINLESS STEEL CORE TUBE, PLUGNUT, SPRINGS, AND PISTON.
ENCLOSURE = WATERTIGHT NEMA 4X. CORROSION RESISTANT. IP66. OUTDOOR RATING
- INSULATED AND HEATED ENCLOSURE. "HOT-BOX" OR DISTRICT APPROVED EQUIVALENT.
- 1" FLOW METER PER ELECTRICAL DESIGN REQUIREMENTS.
- 3/4" UNION. DEMONSTRATE REMOVAL AND RE-ATTACHMENT.
- PIPING TO BE SCH 80 PVC PIPE, 1 INCH DIAMETER.
- 4" THICK, 3,000 PSI, CONCRETE SLAB. 58"Lx24"Wx4" MIN. THICKNESS.
- SUPPORTS PER PIPE SUPPORT DETAIL C7.2
- FILL ANNULAR SPACE WITH DOW CORNING 732 MULTI-PURPOSE SEALANT. JOINT SHALL BE WATERTIGHT.
- 3/4" PVC MANUAL BALL VALVE
- CHECK VALVE, 1.25 INCH, SPEARS MODEL NO. S1780 WITH EPDM SEALS, 150 PSI PRESSURE RATING, STAINLESS STEEL SPRING ASSISTED OPERATION. SELF-CLOSING STYLE DESIGNED FOR HORIZONTAL INSTALLATION/OPERATION.
- FLOW CONTROL ASSEMBLY. ORENCO 1.25 INCH MODEL FCA. INCLUDE SPARE ORIFICE PLATES WITH THE FOLLOWING ORIFICE SIZES (IN INCHES): 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90 0.95, 1.00 AND 1.05.
- 90 W, 120 V, SINGLE PHASE HEAT TRACE TAPE.
- 1" x 3/4" EXPANSION FITTING



1 HOT BOX #1 - FLOW METER & ISOLATION VALVE
NOT TO SCALE



3 HOT BOX #2 - AIR GAP PIPING
NOT TO SCALE



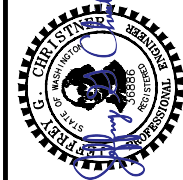
2 HOT BOX #2 - AIR GAP PIPING
NOT TO SCALE

NOTES:

- ALL WORK PERFORMED ID TO MEET MANUFACTURER'S RECOMMENDATIONS, AND MOST RECENT LOCAL AND STATE PLUMBING CODES.
- ALL CORROSIVE SURFACES SHALL BE CLEANED, PRIMED AND PAINTED WITH A MINIMUM OF (2) COATS OF EPOXY PAINT, AS APPROVED BY THE OWNER.
- ALL PIPING INSIDE HOT BOX #1 AND #2 TO BE SCH. 80 PVC PIPE

NO.	REVISIONS	BY	DATE

11-02-2022



DESIGNED BY	JCC
DRAWN BY	JGS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY
WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

CIVIL DETAILS

SHEET	DATE	SCALE	JOB NUMBER
C7.6	NOV 2, 2022	AS SHOWN	2020-070
PAGE	48	OF	96



BID SET